

INTERPRETING DATA IN BIOLOGY

Book 2

Peter Ampt

June Keys



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Peter Ampt and June Keys



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Preface

The purpose of this series of books is to provide opportunities for senior secondary students to develop skills in manipulating, reading, describing, analysing and interpreting biological data. In many cases the data is from actual historical experiments from scientific literature and in other cases has been generated or approximated from real data.

Teachers will find the activities in this book useful as supplementary material for practical work, as materials for relief teachers or as homework. Suggested answers for all questions are provided.

This book contains 17 stand-alone activities. Every activity should take approximately 40 minutes to one hour to complete. Each activity requires students to:

- 1** Read the data, which is presented as tables, graphs, flow charts, diagrams or text.
- 2** Translate or manipulate the data into a more convenient form.
- 3** Describe in words the trends in the data using a scaffold example as a guide.
- 4** Analyse or interpret the data and relate it to existing theoretical knowledge.

As students become familiar with the format, they will become more able to complete the activities with minimal assistance. Less-able students will usually be able to complete each activity up to Step 3. Many students may require mediation by the teacher to complete the final stage of each activity.

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The effects of temperature and pH on enzyme activity

Introduction

Enzymes are proteins that act as catalysts for the thousands of chemical reactions that occur in living things. Without enzymes, these chemical reactions would not happen in the coordinated way needed to sustain life. There are a large number of enzymes in every cell of every living thing, and each enzyme works on a specific chemical reaction.

Each enzyme has an optimum range of conditions (which include temperature and pH) at which its activity is greatest. The activity of the enzyme will be much less outside the optimum range. The optimum range depends on the type of organism and the part of the organism in which the enzyme is found.

In animals, a number of enzymes are involved in digestion. Two examples are:

- **Rennin**—an enzyme present in the stomachs of young mammals that causes milk solids (called curds) to separate from the liquid in milk (called whey). This separation into curds and whey is the first step in the digestion of milk. Junket tablets contain rennin.
- **Pepsin** is an enzyme in the stomach of many mammals that begins the breakdown of protein.

Aim

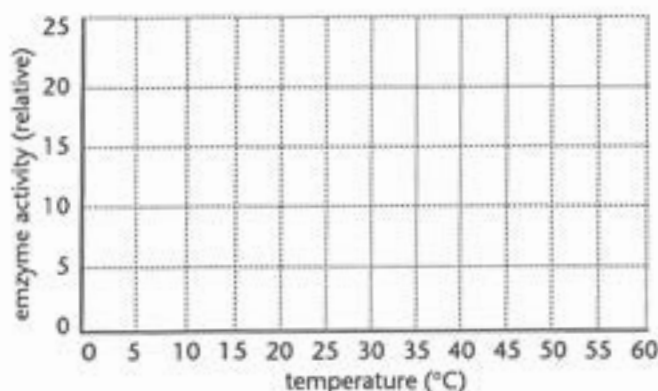
- 1 To compare the effect of temperature on the enzymes of endothermic and ectothermic animals.
- 2 To investigate the effect of temperature on the activity of the digestive enzyme rennin.
- 3 To investigate the effect of pH on the activity of the digestive enzyme pepsin.

Data

PART 1 THE EFFECT OF TEMPERATURE ON THE ACTIVITY OF THE ENZYMES OF ENDOTHERMIC AND ECTOTHERMIC ANIMALS

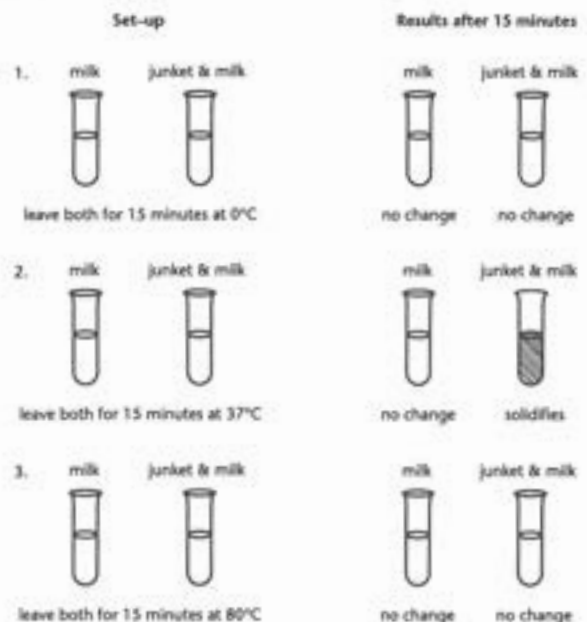
Temperature (°C)	0	5	10	15	20	25	30	35	40	45	50	55	60
Activity of a mammalian (endothermic) enzyme	0	0	1	2	3	6	12	25	14	4	2	1	0
Activity of a reptile (ectothermic) enzyme	0	1	4	8	15	22	25	22	14	9	3	1	0

Draw a line graph of the data on the grid below.



PART 2 INVESTIGATING THE EFFECT OF TEMPERATURE ON ENZYME ACTIVITY

The diagram on the right shows an experiment on the effect of rennin on milk at different temperatures.



Complete the results table below.

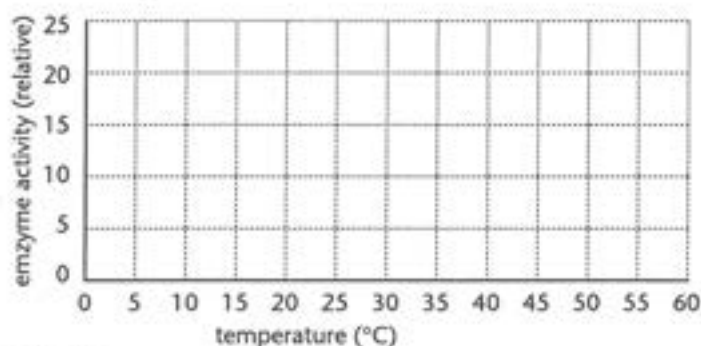
Treatment temperature (°C)	Milk only	Milk and rennin
0		
37		
80		

PART 3 INVESTIGATING THE EFFECT OF pH ON ENZYME ACTIVITY

Enzyme activity	0	0.5	8	10	8	0.5	0	0	0	0	0	0
pH	1.5	2	3	3.5	4	5	6	7	8	9	10	11

Note: pH < 7 is acidic, pH = 7 is neutral pH > 7 is basic or alkaline.

Draw a line graph of the data on the grid below.



Interpreting the data

PART 1

- 1 Describe the effect of temperature on enzyme activity in mammals.
- 2 In mammals, at what temperatures is enzyme activity above a value of 10?
- 3 Describe the effect of temperature on enzyme activity in reptiles.
- 4 In reptiles, at what temperatures is enzyme activity above a value of 10?
- 5 Why do you think the optimum temperature range for enzymes in mammals is different from enzymes in reptiles?

PART 2

- 6 What has caused the milk to solidify?
- 7 Why is the test-tube containing 'milk only' included in the experiment?
- 8 What factors have been varied in this experiment?
- 9 What factors have been kept constant?
- 10 At what temperature is the rennin active?
- 11 Would you expect the digestive enzymes of reptiles to also be active at this temperature? Explain your answer.

PART 3

- 12 Describe the effect of pH on the activity of pepsin.
- 13 At which pH did pepsin work best?
- 14 The digestive enzymes in the small intestine have an optimum pH level of between 5 and 7. Draw another line on the graph to show the activity of these enzymes. Explain why they have a different optimum pH to pepsin.

Feedback control of blood sugar in humans

Introduction

Maintaining a relatively constant blood sugar level is critical to the normal functioning of humans and most other multicellular animals. Blood carries glucose to the cells, where it is used for cellular respiration. The human body is able to achieve a consistent supply of glucose to the cells despite a fluctuating supply of glucose from the digestive system and the changing demand for glucose due to different levels of exertion. The mechanism for the regulation of blood glucose is a type of feedback control. This involves the hormones insulin and glucagon, the pancreas and the liver.

Some of the important words in the control of blood sugar are:

- Hypoglycaemia—low levels of blood glucose
- Hyperglycaemia—high levels of blood glucose
- Glycogen—substance into which glucose can be converted for storage
- Gluconeogenesis—production of glucose from proteins and/or fats
- Glycogenolysis—breakdown of glycogen to form glucose
- Glycogenesis—production of glycogen

Aim

To interpret diagrams to describe the control of blood sugar in humans and develop an understanding of a feedback mechanism.

Data

FIGURE 1: THE MOVEMENT OF BLOOD THROUGH THE PANCREAS AND LIVER.

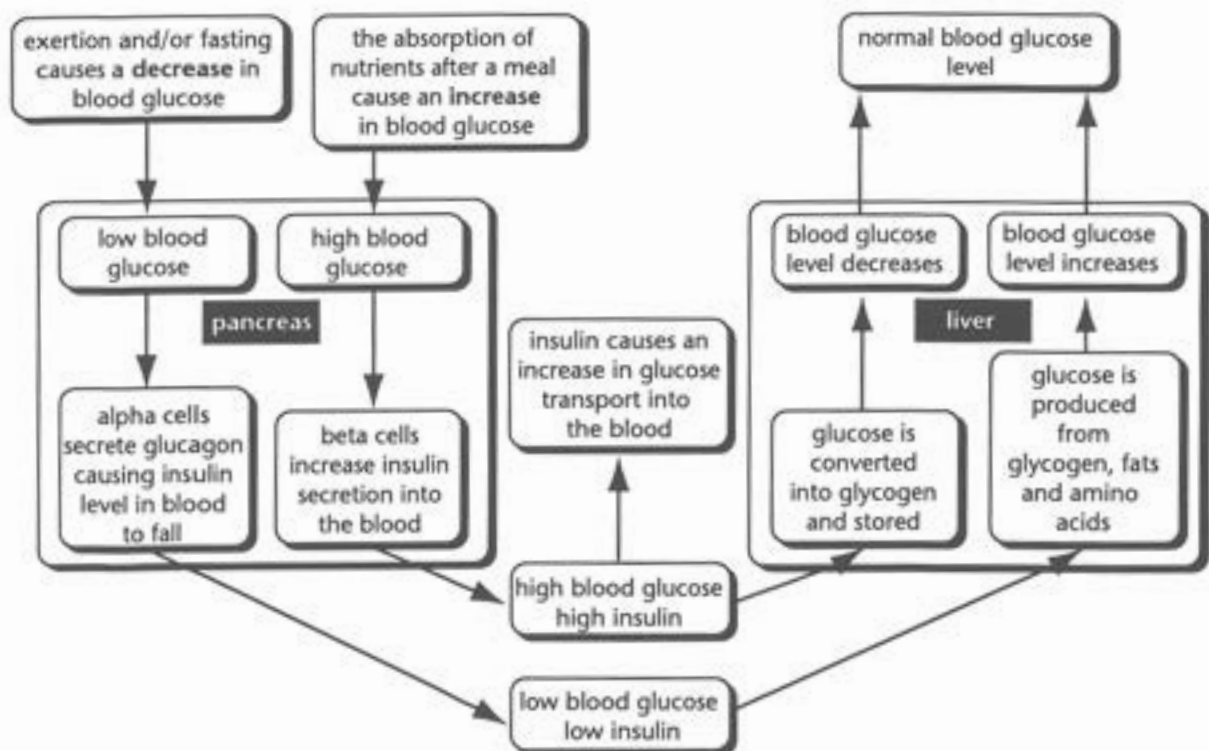
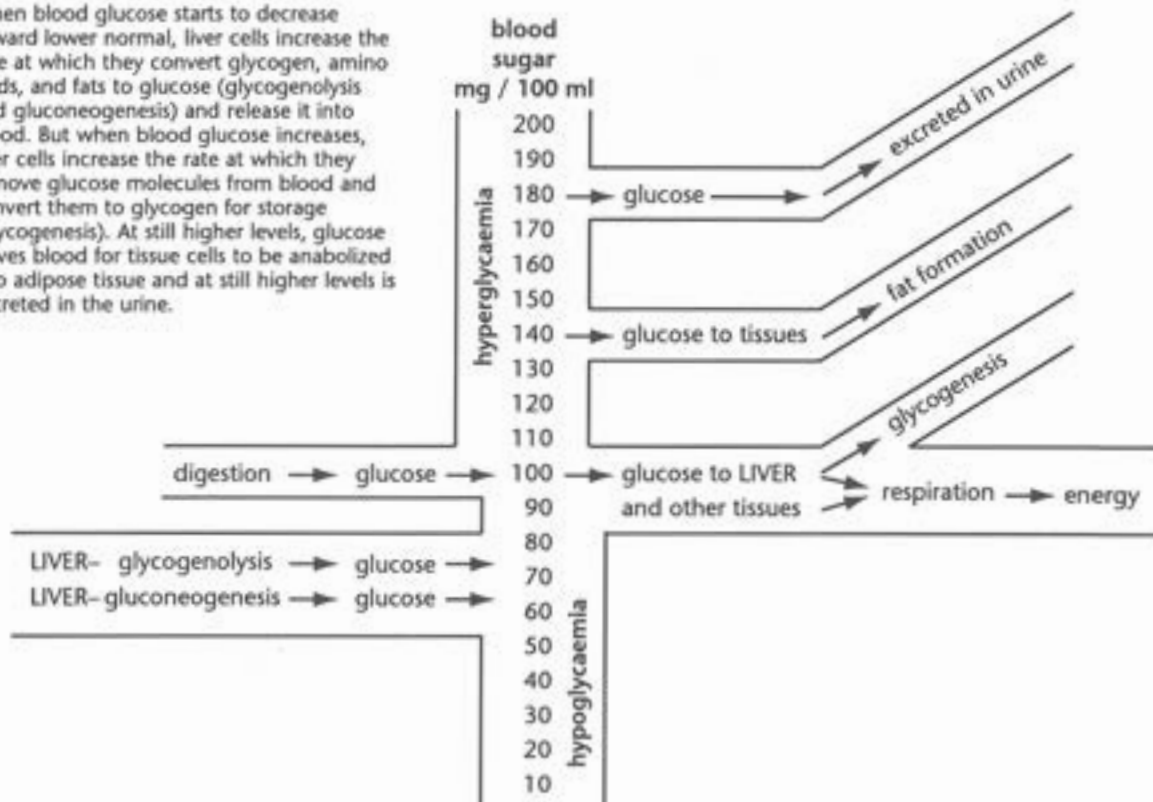


FIGURE 2: EFFECT OF BLOOD SUGAR LEVEL ON GLUCOSE METABOLISM

When blood glucose starts to decrease toward lower normal, liver cells increase the rate at which they convert glycogen, amino acids, and fats to glucose (glycogenolysis and gluconeogenesis) and release it into blood. But when blood glucose increases, liver cells increase the rate at which they remove glucose molecules from blood and convert them to glycogen for storage (glycogenesis). At still higher levels, glucose leaves blood for tissue cells to be anabolized into adipose tissue and at still higher levels is excreted in the urine.

**Reference**

Anthony, C.P. and Kolthoff, N. J. *Textbook of anatomy and physiology*, The C V Mosby Company, St Louis, 1975.

Interpreting the data

- 1 What can cause the blood entering the pancreas to have a low level of glucose? Explain.
- 2 How does the pancreas respond to low levels of blood glucose?
- 3 How does the liver respond to low levels of insulin and glucose in the blood?
- 4 What can cause the blood entering the pancreas to have a high level of glucose? Explain.
- 5 How does the pancreas respond to high levels of blood glucose?
- 6 What effect does insulin have on glucose transport into the cells?
- 7 How does the liver respond to high insulin and high glucose levels in the blood?
- 8 At what blood sugar level is glucose excreted in urine?
- 9 What happens to glucose when blood sugar drops below 80mg/100ml?
- 10 List three ways that glucose can be added to the blood.
- 11 List four ways that glucose can be removed from the blood.

Making links

- 12 What stimulates insulin production?
- 13 What are the effects of insulin on blood glucose?
- 14 What stimulates glucagon production?
- 15 What is the effect of glucagon on blood glucose?
- 16 Why do you think the control of blood sugar levels is called a feedback control?

Products extracted from donated blood

Introduction

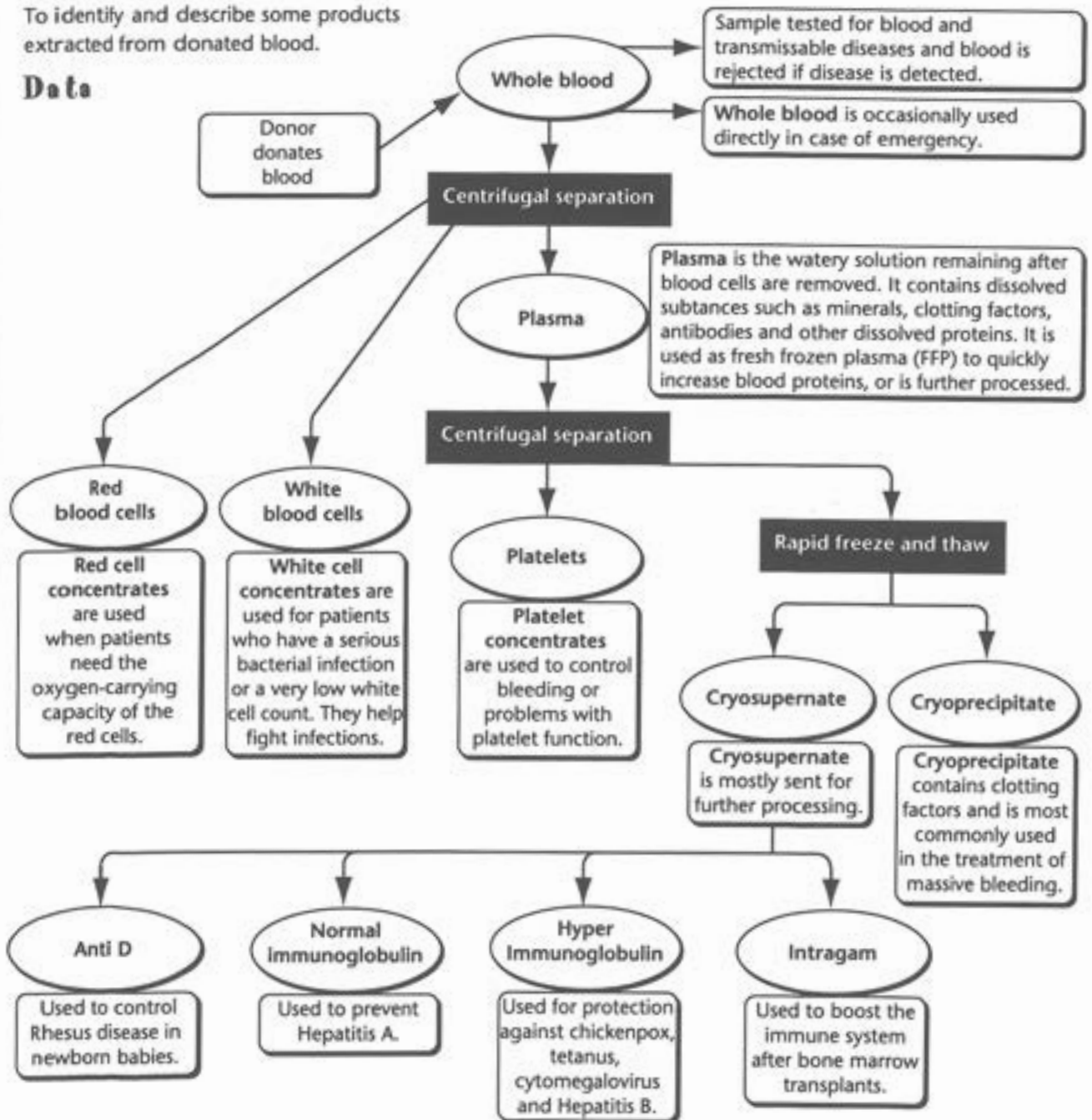
Blood is a complex mixture of cells, water and dissolved substances. When people donate blood, it may occasionally be used as whole blood if it is needed urgently. However, it is usually separated into its components and then made into a range of blood products. As a result, maximum use is made of every litre of blood that is donated. In fact, a single donation can be used to treat several patients suffering from different illnesses. This technique is known as 'blood component therapy'.

The process of separation involves a number of stages. First, a centrifuge is used to separate the cells from the water solution (plasma). Then, the dissolved substances are removed from the plasma and converted into a range of useful products.

Aim

To identify and describe some products extracted from donated blood.

Data



Interpreting the data

- 1 Explain why some donated blood is rejected.
- 2 Under what circumstances is whole blood used?
- 3 What components of whole blood are separated at the first centrifuge?
- 4 What are the uses of red and white blood cell concentrates?
- 5 Which component of whole blood is used to produce Intragam?
- 6 What extracted products from donated blood are used for:
 - a treating massive bleeding?
 - b patients who have a low white cell count?
 - c controlling bleeding, following a severe haemorrhage?
 - d managing Hepatitis B?
 - e protection against chicken pox?
 - f boosting the immune system after a bone marrow transplant?
- 7 What blood products are extracted from blood plasma?
- 8 What process is used to separate cryosupernate and cryoprecipitate from plasma?
- 9 What products are extracted from cryosupernate?
- 10 What is the function of red blood cells?
- 11 Draw lines to link each blood product below with its use. One is done for you.

Blood product	Use
Whole blood	To boost the immune system after bone marrow transplants
Plasma	To control Rhesus disease in newborn babies
Red blood cells	In cases of serious infection or due to a very low white cell count
White blood cells	To prevent Hepatitis A
Cryoprecipitate	To increase of the oxygen-carrying capacity of the blood for example, after bleeding
Hyper Immunoglobulin	When there is severe blood loss and patient will die without it
Intragam	To control bleeding due to deficiencies in the clotting of blood
Normal Immunoglobulin	For protection against a variety of viral diseases
Anti D	Most commonly used in the treatment of massive bleeding
Platelets	To quickly increase blood proteins

Temperature regulation in ectotherms and endotherms

Introduction

The body temperature of ectothermic ('warmed from the outside') animals is determined by the temperature of their environment. Ectotherms are unable to maintain the same level of activity when the environmental temperature changes. In extreme cold, for example, ectotherms become too cold to move and, as the environment warms their bodies, they become more active. In contrast, endothermic ('warmed from the inside') organisms generally maintain a constant body temperature despite changes in environmental temperature. Being endothermic enables animals to maintain activity in a wide range of environmental temperatures.

Animals that live in deserts experience extreme daily variations in temperature—from below zero in the early morning up to more than 40°C in the afternoon. Animals that live high in alpine areas experience months of extremely low temperatures followed by months of more moderate temperatures. Even in summer, temperatures often drop below freezing at nights.

Animals usually have several adaptations to survive in the range of temperatures they experience in their habitats. These adaptations may be structural (features of the structure of the body), behavioural (specific behaviours) or physiological (features of the inner workings of the body). The adaptations for the regulation of body temperature in endotherms differ greatly to those in ectotherms. In order to maintain a constant body temperature, endothermic animals have a number of adaptations to help them retain heat when it is cold or lose heat when it is hot. Ectotherms often have adaptations that help them to maximise the beneficial and minimise the adverse effects of changes in environmental temperatures.

Aim

To use secondary data to identify adaptations that help regulate the body temperature of ectothermic and endothermic animals from desert and alpine environments.

Data

Thorny Devil—a small (about 9 cm) animal that lives in the sand plains, spinifex desert and arid scrub of Central and Western Australia. It has a short, slightly flattened body covered by a thick scaly skin with conical spines along its back and head. It lies in the sun with its body flattened against the warm sand in the morning and lifts its body off the hot sand in the middle of the day. It remains motionless for long periods of time throughout the day and is often found in the shade or buried in the sand or a burrow. It mostly eats ants and can make use of dew by absorbing water directly through its skin.

Mitchell's Hopping Mouse—a small (about 5 cm) animal that lives in arid areas of Central Australia. It has a small, slim body covered with thick fur, long, thin legs and large, thin ears that have a dense network of blood vessels and sweat glands. It is usually active from the late afternoon, through the night until mid-morning. On very cold nights it fluffs out its fur. It remains in its burrow from late in the morning until late in the afternoon.

Corroboree Frog—a small (1–3 cm) animal that lives in sphagnum moss in high altitude areas of southern Australia. It has a small, thin body covered with a wet, slimy skin. It is coloured black with bright yellow features. It stays in the moss overnight and emerges early in the morning to bask in the sun. It is only active on warm days in summer and feeds on small insects and other arthropods to accumulate fat tissue for the winter. It spends the winter burrowed in moss under the snow. In late winter, its sex organs develop so that when it first becomes active in spring, it is ready to mate. After mating, it begins feeding again.

Mountain Pigmy Possum—a small (about 5 cm) animal that lives in the alpine areas of south-eastern Australia above 1400 m. It has thick fur, a round stocky body, short legs and a long, prehensile tail. It has small ears that have limited blood circulation. It feeds on insects, other arthropods, seeds and fruits throughout spring, summer and autumn and often doubles its body weight. In winter, it hibernates and rolls itself into a furry ball, tucking its nose, ears, tail and feet into its body. It also slows down its metabolic rate.

Recording the data

Extract the information about adaptations for heat control from the data above and record it in the table. Some are done for you.

Animal	Environment	Adaptations for heat control		
		Structural	Behavioural	Physiological
Thorny Devil	Desert			
Mitchell's Hopping Mouse	Desert			
Corroboree Frog	Alpine			
Mountain Pigmy Possum	Alpine			

Interpreting the data

- Which of the animals in the table are endothermic? On what evidence did you base your answer?
- Which of the animals in the table are ectothermic? On what evidence did you base your answer?
- Explain why the Thorny Devil behaves differently in the morning than in the middle of the day.
- What adaptations help the Mitchell's Hopping Mouse to maintain its body temperature throughout the day?
- What adaptations help the Mitchell's Hopping Mouse to maintain its body temperature throughout the night?
- Explain why the Corroboree Frog is less active at night and in winter.
- What is the advantage to the Mountain Pigmy Possum of rolling itself into a furry ball while it hibernates?
- Contrast the structural adaptations of Mitchell's Hopping Mouse and the Mountain Pigmy Possum and discuss why they are adapted to their environments.
- Why do both the Thorny Devil and the Corroboree Frog bask in the sun in the morning?
- Contrast the body temperatures of each of the animals over a 24-hour period.

Making links

Draw a line to link the adaptation with the reason or advantage. One has been done for you.

Adaptation	Reason or advantage
Large thin ears	Radiate heat into surrounding air
Lizard lying in the ground in the early morning	Avoid extremes in temperatures
Roll into a furry ball	To warm the body
Thick fur	To insulate, keeping heat inside body
Lizard burrowing	To prevent heat loss by reducing the SA/Vol

Regulation of body temperature in humans

Introduction

In endothermic animals such as mammals and birds, normal metabolism requires that the body remain at a relatively constant temperature. In order to do this, the body must achieve a balance between heat loss and heat production by the body. If the body becomes too cold it will attempt to minimise heat loss at the same time as it increases heat production. If the body becomes too hot it will attempt to minimise heat production and maximise heat loss.

Aim

To describe how humans respond to changes in temperature and relate these responses to heat production and heat loss.

Data

TABLE 1: HUMAN RESPONSES TO ENVIRONMENTAL TEMPERATURE

Environ. temp (°C)	Body temp (°C)	Observed responses	Observed behaviour
0	21	finger nails, toenails, lips and nose appear blue, sleepiness and disorientation, coma and death	
10	35	skin feels cold to the touch, fingers, toes, nose and ears cold, lips appear blue, violent shivering	curled in a ball
20	37	skin feels cool to the touch, muscle tension, slight to moderate shivering	arms crossed, hunched shoulders, rubbing skin, swinging arms, moving around
30	37	slight perspiration	appears comfortable and relaxed, moving freely
40	37	perspiration, redness in the face and skin	sitting still, arms and legs apart, fanning
50	38	increased perspiration, redness spreading to body	sitting still, arms and legs spread out
60	40	profuse perspiration, redness over whole body, dizziness	lying down, arms and legs spread out
70	48	sweating diminishes, coma and death	collapse

TABLE 2: THE INFLUENCE OF HEAD TEMPERATURE ON HEAT PRODUCTION AND HEAT LOSS IN HUMANS

Head temperature (°C)	36.4	36.5	36.6	36.7	36.8	36.9	37.0	37.1	37.2	37.3	37.4	37.5	37.6
Heat production (joules/second)	336	320	294	275	252	205	147	86	86	86	86	86	86
Evaporative heat loss (joules/second)	8	8	8	8	8	8	8	95	190	245	300	345	390

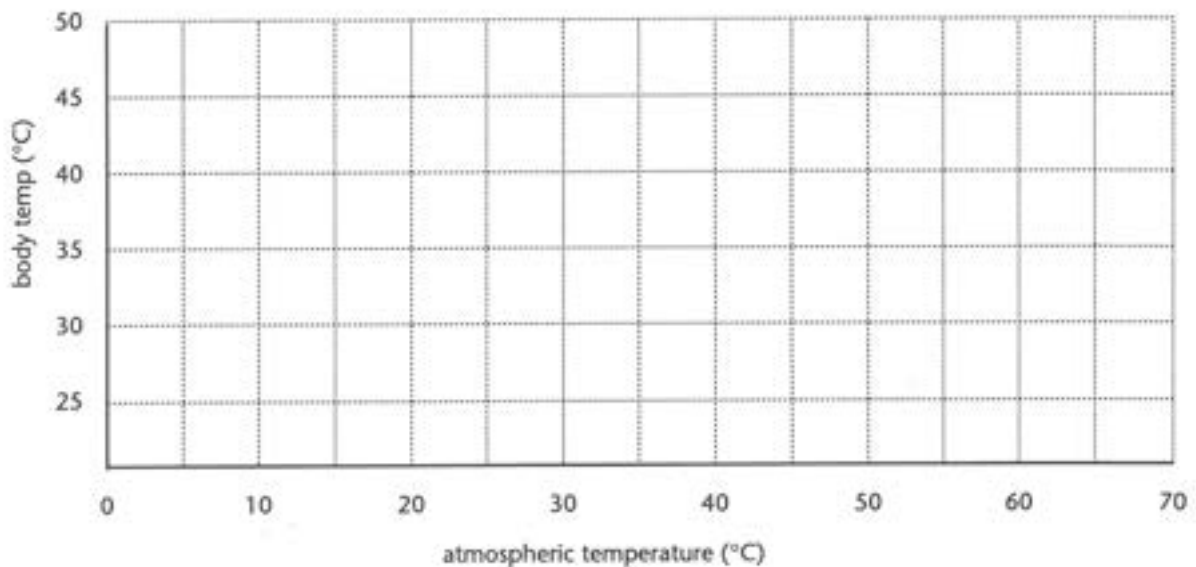
Reference

Guyton, A. C. *Textbook of medical physiology*, W. B. Saunders Company, Philadelphia, 1976.

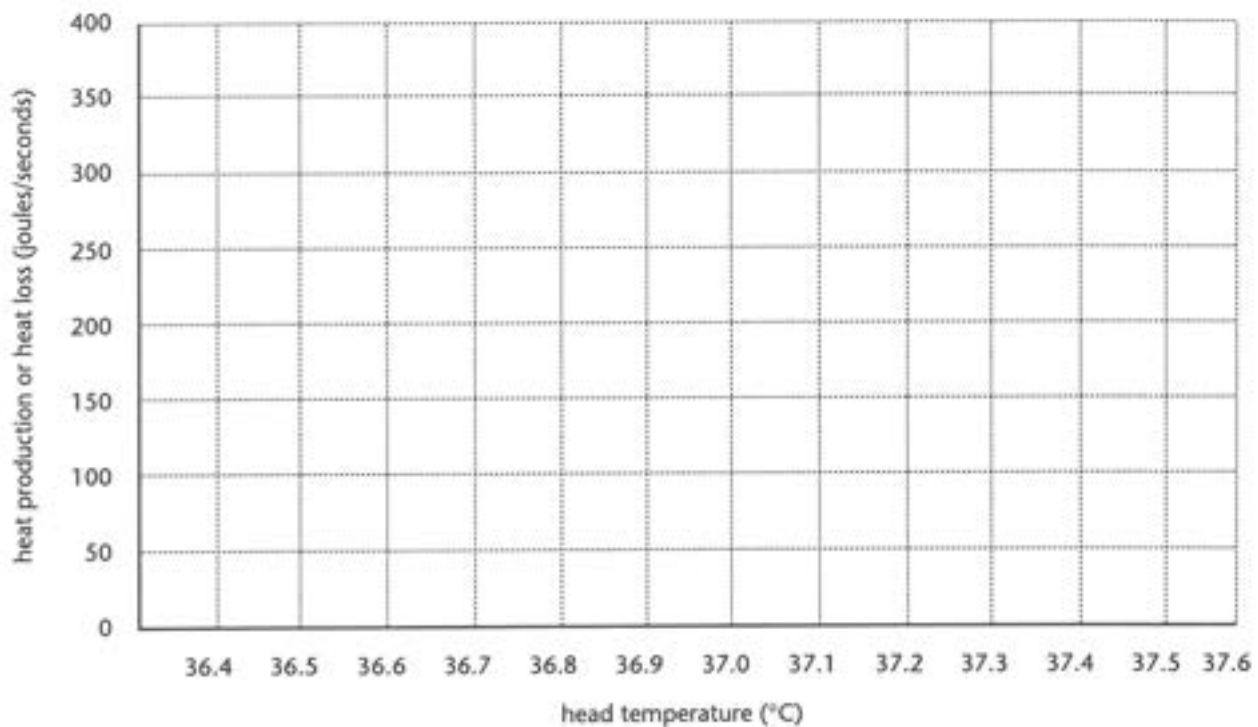
Graphing the data

1 Graph the data in Tables 1 and 2 on the axes below

BODY TEMPERATURE AT DIFFERENT ATMOSPHERIC TEMPERATURES



HEAT PRODUCTION AND HEAT LOSS AGAINST HEAD TEMPERATURE



Describing the data

- 1** Between what atmospheric temperatures is the body able to maintain a relatively stable body temperature?
- 2** What are the two main body responses to environmental temperatures over 30°C?
- 3** What are the three main body responses to prolonged exposure to temperatures below 20°C?
- 4** Between what head temperatures does the body switch from heat production to heat loss?

Interpreting the data

- 5** Link each of the body responses and behaviours below to a physiological explanation. One has been done for you. *Please note:* in some cases, more than one body response or behaviour can have the same physiological explanation.

Body response or behaviour	Physiological explanation
Sleepiness and disorientation	
Curled in a ball, arms crossed	These behaviours reduce the surface area of the body exposed to cold and so reduce the rate of heat loss.
Lips and nose appear blue	These behaviours increase the surface area of the body and so increase the rate of heat loss.
Swinging arms, moving around	Person has hypothermia and their low body temperature is now disrupting the biochemistry of the brain.
Skin feels cold	Perspiration evaporates from the skin causing evaporative cooling. Heat moves from the blood into the cool skin.
Shivering	Peripheral blood vessels (those in the skin) are constricted (closed up) reducing the blood supply to the skin. This cools the skin and reduces the rate of heat loss.
Perspiration	Peripheral blood vessels are dilated (opened up) increasing the blood supply to the skin, increasing heat loss.
Arms and legs spread out	Due to prolonged exposure to heat, the body has become dehydrated and, as a result, is no longer able to perspire at the same rate.
Redness in the face and skin	Body movement increases heat production by the body.
Sweating diminishes	

- 6** Explain how behaviour, perspiration and the dilation of peripheral blood vessels work together to maintain a constant body temperature at high environmental temperatures.
- 7** Explain how behaviour, increased metabolic rate and constriction of peripheral blood vessels work together to maintain a constant body temperature at low environmental temperatures.
- 8** Describe the evidence in the data that supports the hypothesis that the control of body temperature is located in the brain.

Temperature control and water loss in humans

Introduction

Humans, like birds and other mammals, are endothermic. The human body is maintained at a relatively constant temperature by a number of mechanisms:

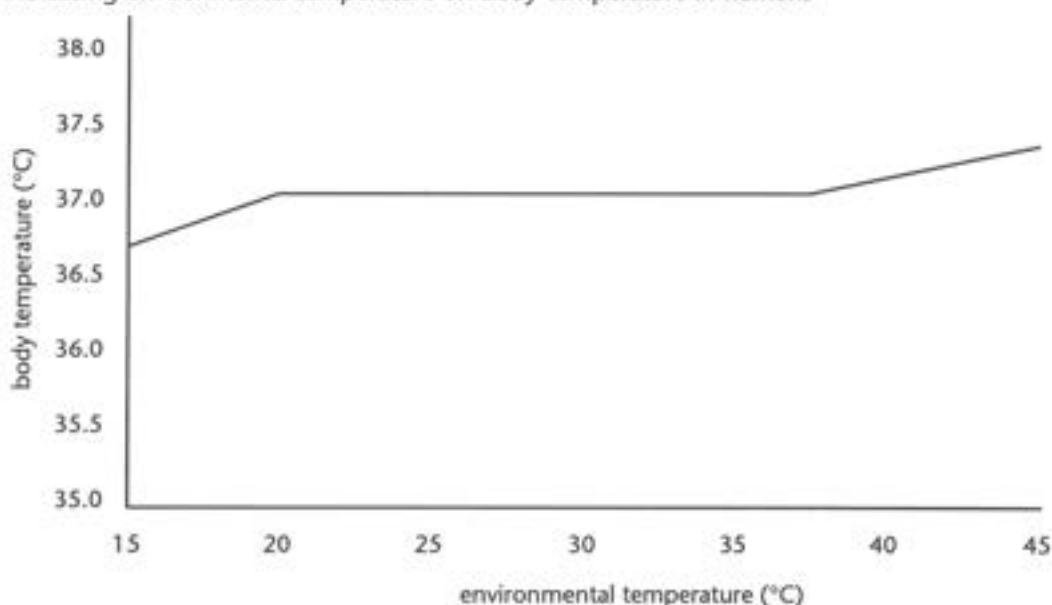
- If the core temperature drops, the body can limit heat loss by constricting (narrowing) the peripheral blood vessels (in the skin), raising body hair and generating extra heat by shivering.
- If the core temperature rises, the body can lose heat by sweating. As the sweat evaporates, it cools the skin and the blood flowing through the skin. This process is made more efficient by dilating (widening) the peripheral blood vessels so that more blood comes in contact with the cooled skin layer.

Aim

To investigate the mechanisms for temperature control in humans and the loss of water through sweating.

Data

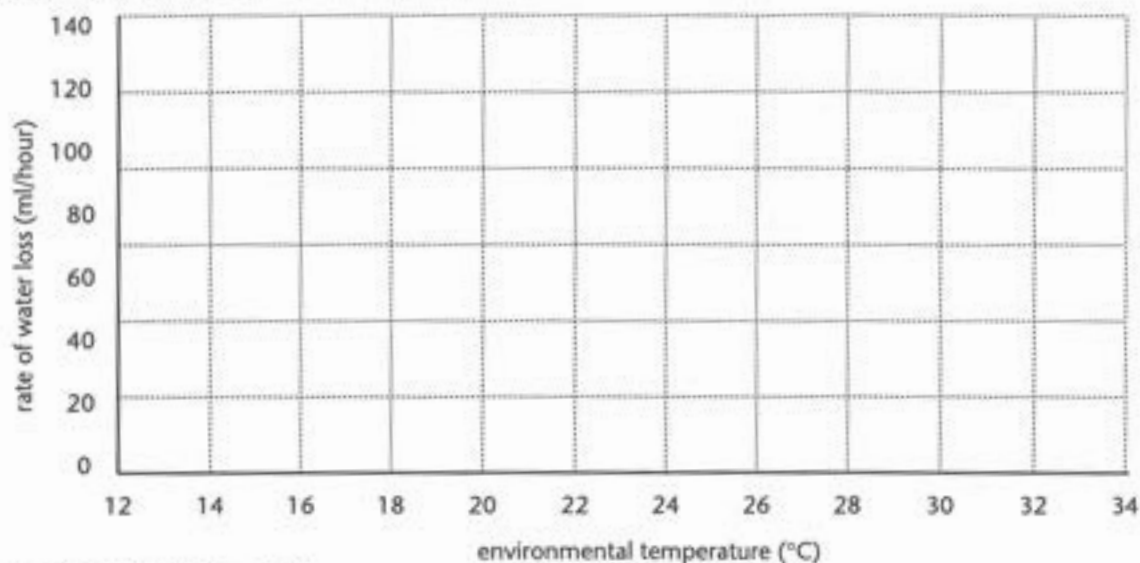
Effect of increasing environmental temperature on body temperature in humans



RATE OF WATER LOSS IN HUMANS DURING INCREASING ENVIRONMENTAL TEMPERATURE

Environmental temperature (°C)	Rate of water loss (ml/hour)	Observations
12	20	constant shivering
14	22	extremities cold
16	24	shivering
18	26	skin cold to touch
20	28	goose bumps appear on the skin.
22	30	
24	32	
26	34	
28	36	sweating
30	40	thirsty
32	68	skin colour becomes redder
34	118	lower volume of darker coloured urine.

Graph the environmental temperature and the rate of water loss below



Describing the data

- 1 Describe the relationship between body temperature and environmental temperature.
- 2 Describe the relationship between water loss and environmental temperature.
- 3 At what temperature does water loss due to sweating begin?
- 4 What other effects does increasing environmental temperature have on the body?

Making links

The observations in the table above show how the body responds to water loss by sweating

- 5 Draw lines to link the body's response with the organs involved and the explanation. Two are done for you.

BODY RESPONSES TO CHANGE IN ENVIRONMENTAL TEMPERATURE

Body's response	Organs involved	Explanation
shivering	kidneys	reduced blood volume and increased blood concentration → hormone signal → kidneys reabsorb more water → less water in urine → urine more concentrated → less water lost
skin cold to touch	brain and peripheral blood vessels	decrease in body temperature → brain signals an increase in muscle tension → increasing muscle tension results in shivering → heat production increased by up to 500%
feeling thirsty	brain	increase in body temperature → brain signals to blood vessels → blood vessels in skin dilate (widen) → more blood flows to skin → blood cooled → body temperature returns to normal
redder skin colour	brain and peripheral blood vessels	blood more concentrated → thirst centre in brain activated → person feels thirsty and drinks → blood returns to normal concentration
lower volume of darker urine	brain and body muscles	decrease in body temperature → brain signals to blood vessels → blood vessels in skin constrict (become narrower) → less blood flows to skin → heat loss reduced

Write a brief summary of:

- the effect of the peripheral blood vessels on temperature control.
- the effects of sweating on temperature control and water loss.

The adaptations of mangroves

Introduction

Mangroves are trees that form woody plant communities in shallow tidal water. They live on sheltered shores in bays, estuaries and the sheltered side of offshore islands. They make an essential contribution to the ecology of coastal and marine environments. Their roots provide habitats for many marine animals such as crabs, prawns, barnacles, isopods and amphipods, as well as helping to trap nutrient-rich silts and preventing erosion of the bank. Their canopy provides a habitat for about 35 species of insect and many birds that eat mangrove nectar and pollen.

Mangroves are specifically adapted to their tidal environment. This environment includes soil (mud) that is always waterlogged with salty water and a varying water level caused by tides ebbing and flowing underneath the mangroves. These conditions would kill other land plants. So how do mangroves survive?

Aim

To describe the adaptations of mangroves to their tidal environment and explain how these adaptations help mangroves to survive.

Data

A COMPARISON OF THE FEATURES OF THE BRUSH BOX (A COMMON STREET TREE) AND THE GREY MANGROVE AND THEIR HABITATS

	Brush Box	Grey Mangrove
Soil oxygen (by volume)	2 %	<0.1 %
Root system	All below ground	Below and above ground, including pneumatophores* with lenticels**
Fruits and seeds	Fruits split open while still on the tree to release many seeds which are spread by birds.	Fruits contain a single seed which germinates inside the fruit while still on the tree.
Seedlings	Seeds germinate and grow into seedlings in the soil if conditions are suitable.	Seedling germinates while still on the tree, then falls and floats in the water. The root system develops quickly when the seedling touches the mud, anchoring the young plant before the next tide.
Relative salt concentration in soil water	▲	▲▲▲▲▲
Relative salt concentration in root parenchyma cells	▲▲	▲▲▲▲▲
Relative salt concentration in xylem	▲▲	▲▲▲▲▲
Relative salt concentration in leaf cells	▲▲▲	▲▲▲▲▲
Relative salt concentration in oldest leaves	▲▲▲	▲▲▲▲▲
Salt deposits on leaf surfaces	No	Yes

▲ represents salt mounds.

* pneumatophores (peg roots) are above ground roots that grow straight up

** lenticels are structures at the tips of pneumatophores that absorb oxygen directly from the air

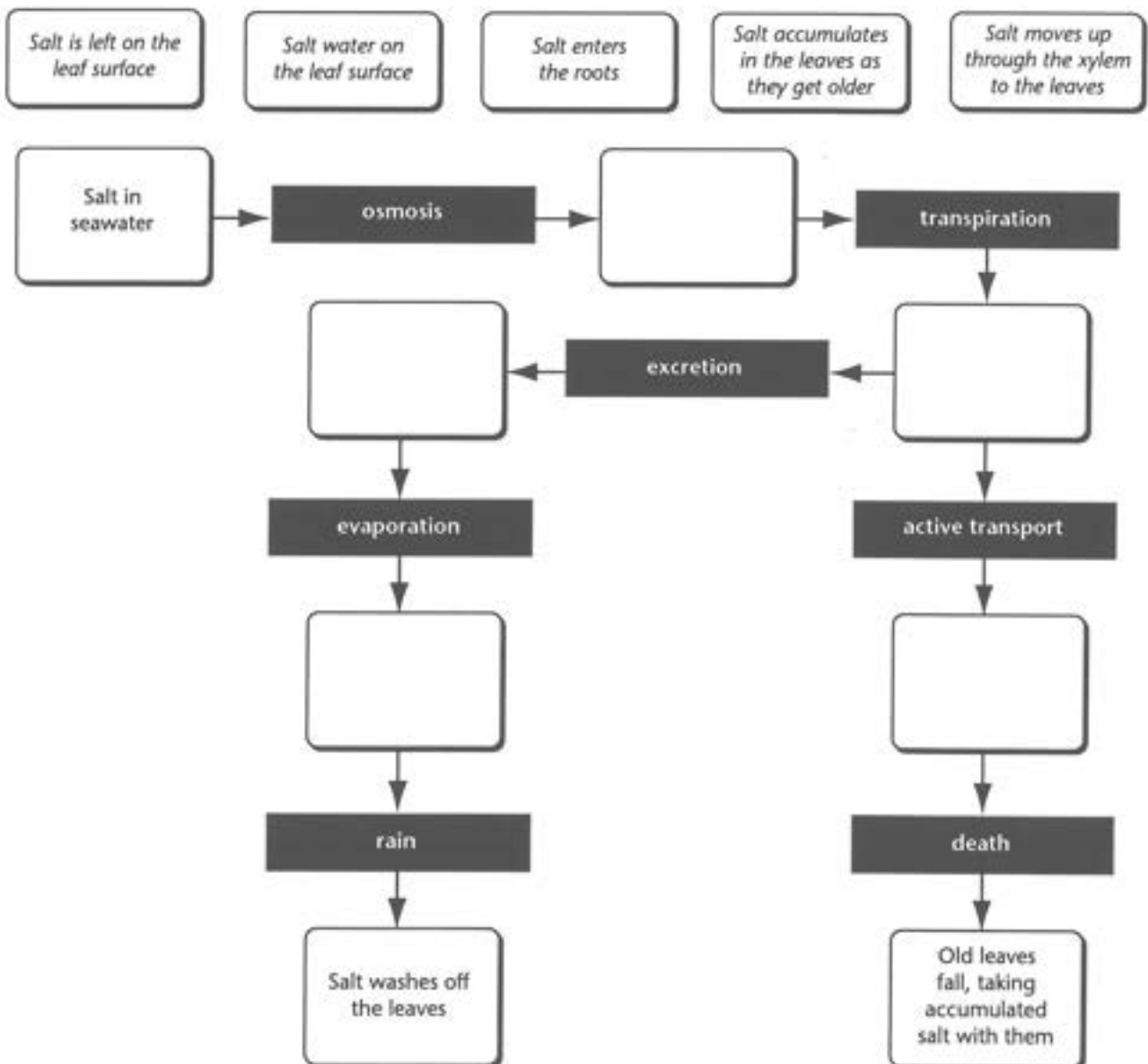
Reference

Blaxland, B. Mangroves, Australian Museum, Sydney, 2000.

Interpreting the data

- 1 Why do plant roots need oxygen?
- 2 How do the roots of terrestrial (land) plants obtain oxygen?
- 3 What adaptations do the roots of Grey Mangroves have that help them obtain oxygen?
- 4 What conditions are required for terrestrial plants to germinate and where does this usually occur?
- 5 In what ways do the conditions for the germination of Grey Mangroves differ from terrestrial plants?
- 6 What adaptations do mangroves have to overcome these conditions for germination?
- 7 Water moves from the soil into plant roots by osmosis. Compare the salt concentration of soil water and root parenchyma for both species and explain how these concentrations promote the absorption of water.
- 8 What would happen to the uptake of water by the roots of a Brush Box tree if the soil water were as salty as for the Grey Mangrove?
- 9 Mangroves take in salt when they absorb water. To survive, they need to get rid of the excess salt. Complete the flow diagram below by choosing the appropriate sentence to write in each box.

THE PREVENTION OF SALT ACCUMULATION IN GREY MANGROVES



Water balance in terrestrial environments

Introduction

Water balance occurs when the amount of water gained by the animal is balanced by the amount of water they lose over a given period of time. Terrestrial animals gain water from their diet (including food and drink) and from the oxidation of carbohydrates, fats and proteins in their cells (such as by cellular respiration). They lose water when it evaporates from their skin and respiratory system and in their faeces and urine.

The amount of available water in terrestrial environments varies greatly. In arid environments such as deserts, where the amount of available water is usually extremely small, animals have a range of adaptations to maintain water balance. Kangaroo rats generally live in arid environments whereas laboratory rats live in a controlled environment.

Aim

To use data to identify adaptations of the Kangaroo rat to a dry environment.

Data

TABLE: COMPONENTS OF WATER BALANCE FOR THE KANGAROO RAT AND LABORATORY RAT FED THE SAME DIET AND UNDER THE SAME ENVIRONMENTAL CONDITIONS

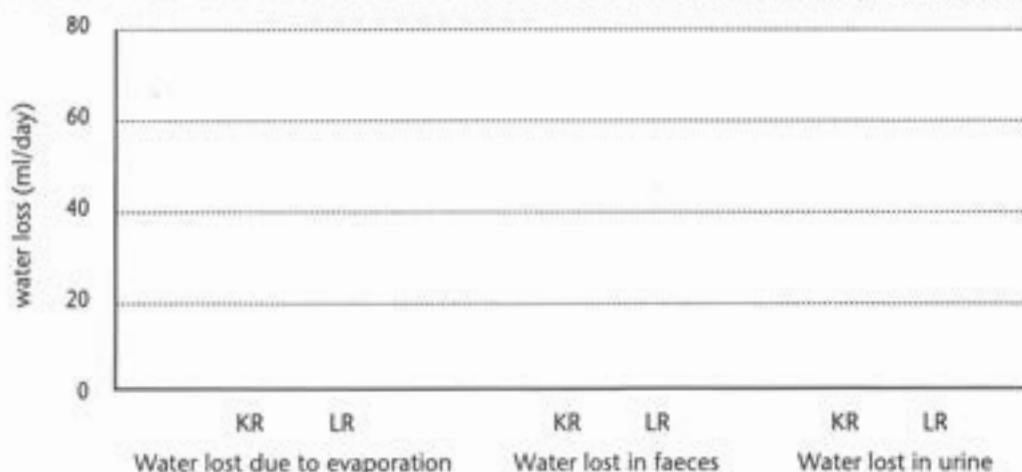
Water loss or gain	ml of water/kg body weight /day	
Description	Kangaroo rat	laboratory rat
Gain due to oxidation of carbohydrate, fat and protein	+54.0 ml	+54.0 ml
Loss due to evaporation	-44.0 ml	-77.0 ml
Loss in faeces	-2.5 ml	-13.6 ml
Loss in urine	-13.5 ml	-22.0 ml
Water balance (excluding diet)		
Minimum water required in diet for water balance		

Reference

Hainsworth *Animal physiology: adaptations in function*, Addison-Wesley Publishing Company, Reading, Massachusetts, 1981

- 1 Complete the table above by calculating the missing values. Note that gains are positive and losses are negative.
- 2 Complete the bar graph below using the data in the table.

BAR GRAPH OF THE COMPONENTS OF WATER LOSS FROM A KANGAROO RAT (KR) AND A LABORATORY RAT (LR)



Interpreting the data

- 1 How is the majority of the water lost from these animals?
- 2 Compare the water lost by the Kangaroo rats and laboratory rats.
- 3 Which of the two animals needs more water in its diet to maintain water balance? Explain.
- 4 Suggest reasons why the water gain due to oxidation is the same for both animals.
- 5 Suggest a behavioural adaptation that could reduce water loss by evaporation.
- 6 Explain how the adaptation you have suggested in question 5 reduces water loss by evaporation.

ADAPTATIONS FOR REDUCING WATER LOSS IN KANGAROO RATS

Draw lines to link the type of water loss with an adaptation for reducing water loss and an explanation of how the adaptation works. There may be more than one adaptation for each type of water loss. One is done for you.

Type of water loss	Adaptation	Explanation
water lost by evaporation	kidneys with very long tubules	reduces water loss through evaporation from the skin
water lost in faeces	tough, oily layer on skin	cools the air so water vapour condenses in the nose before it is breathed out
water lost in urine	long digestive system	absorbs more water from the faeces before it is expelled
	narrow, convoluted passages in the nose	reabsorbs most of the water before it is expelled as urine

- 7 Write sentences explaining how Kangaroo rats reduce water loss:

- due to evaporation
- in faeces
- in urine.

Gill surface area and level of activity in fish

Introduction

Respiration (see box below) is the cellular process by which usable energy (in the form of ATP and NADPH) and heat is produced from the conversion of glucose and oxygen into carbon dioxide and water.



Respiration occurs in almost all organisms, but different organisms respire at very different rates. For example, very active animals require more energy than less active animals and endothermic animals require more energy than ectothermic animals. In both active animals and endothermic animals, a greater rate of respiration is required. Also, since oxygen is needed for respiration, there is a greater demand for oxygen as well.

The oxygen required for respiration is absorbed through moist respiratory surfaces such as gills (in aquatic animals), and lungs (in terrestrial animals). Increasing the surface area of the respiratory surface is one adaptation that allows organisms to meet the demand for oxygen.

Aim

To determine the link between the surface area of the gills of a fish and its level of activity, then compare lungs and gills as respiratory surfaces.

Data

Type of fish	relative gill surface area (per gram of body weight)	habitat/habits	Relative level of activity (1=least active, 5=most active)
 Puffer	505	Swims slowly around in shallow water feeding on corals and small crustaceans.	
 Sea Bass	1111	Leaps into the air when chasing prey in coastal waters but does not swim long distances.	
 Sea Robin	712	Moves slowly around on the bottom of the ocean feeding on crustaceans.	
 Sea Trout	1253	Lives in the open ocean and around the coast. Swims up rivers to breed.	
 Goosefish	51	Moves slowly and rarely, remaining still on the ocean floor until prey comes within reach.	
 Mackerel	2551	Live, travels long distances in the open ocean, swimming very quickly to feed on prey.	

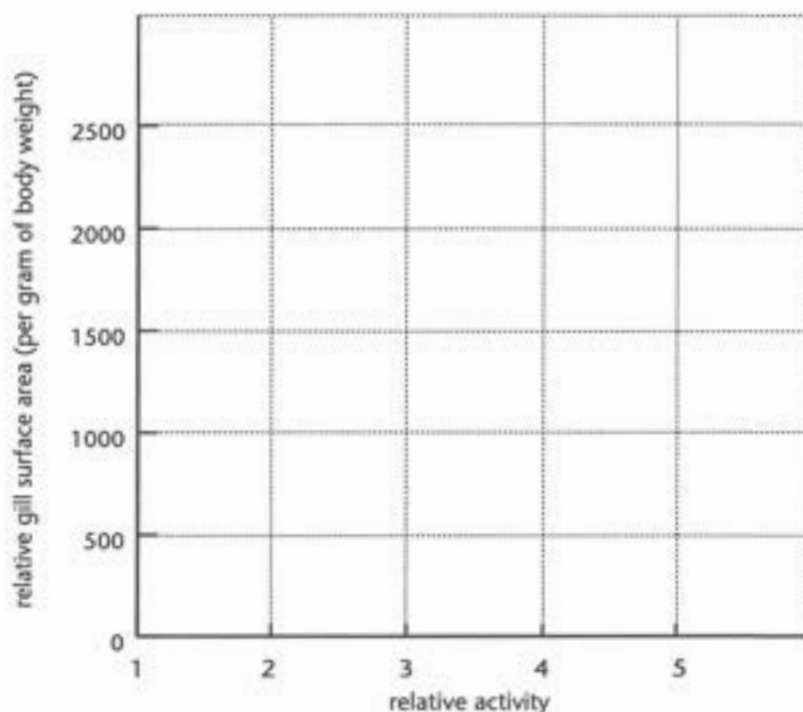
Reference

Hainsworth *Animal physiology: adaptations in function*. Addison-Wesley Publishing Company, Reading, Massachusetts (1981).

Processing the data

- 1** Complete the table by choosing a number from 1-5 to describe the relative activity level of each type of fish.
- 2** Draw a graph in the space below of the relative activity of each fish against their relative gill surface area.

GRAPH OF RELATIVE ACTIVITY AND GILL SURFACE AREA IN FISH



Interpreting the data

- 1** Describe the relationship between gill surface area (per gram of body weight) and level of activity in fishes.
- 2** Explain why there appears to be a link between gill surface area and level of activity in fishes.
- 3** Water temperature may also influence gill surface area (per gram of body weight) in fishes. Describe how and why this factor may influence gill surface area.

Making links

- 4** Would you expect a similar relationship between lung surface area (per gram of body weight) and level of activity in terrestrial animals? Explain.
- 5** Predict whether the lung surface area (per gram of body weight) of endothermic animals would be bigger or smaller than ectothermic animals. Justify your answer.
- 6** Compare the availability of oxygen in water and air. Would you expect lung surface area (per gram of body weight) of terrestrial reptiles (ectothermic animals) to be greater or less than the gill surface area (per gram of body weight) of fish? Explain.
- 7** Make a list of the environmental and biological factors that influence the size of the respiratory surfaces of animals.

Inheritance of eye-colour in *Drosophila*—is it Mendelian?

Introduction

*Drosophila*s are small flies that are often used in the study of genetics. They are suitable for genetic research because they produce a new generation every few hours and have many characteristics caused by mutations that are easy to see. One characteristic is eye-colour. Wild-type *Drosophila* usually have red eyes but some have white eyes.

Mendelian characteristics follow the ratios that Gregor Mendel discovered in his work with peas. For example:

Parents	Type A x Type B
F1	All type A
F2	3 Type A : 1 Type B

Sex-linked characters do not follow Mendelian ratios because these characters are inherited differently in males and females as a result of the gene being located on the X chromosome. Females have two X chromosomes while males have one X and one Y chromosome.

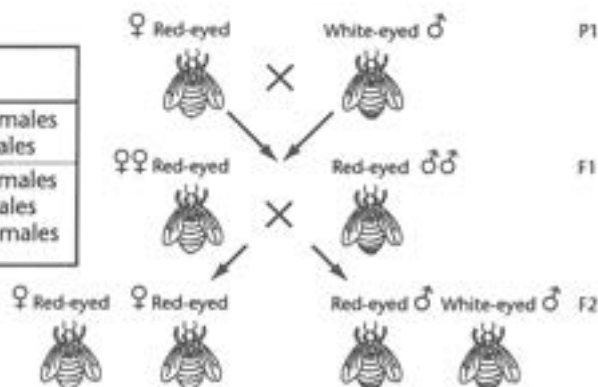
Aim

To use data from crosses of *Drosophila* to work out whether eye-colour is inherited according to Mendelian ratios or is sex-linked.

Data

Cross	Parents	Offspring
Parent 1 (P1)	red-eyed female x white-eyed male	F1 183 red-eyed females 178 red-eyed males
Filial 1 (F1)	red-eyed female x red-eyed male (both offspring from P1 cross)	F2 204 red-eyed females 98 red-eyed males 105 white-eyed males

Reference: Rothwell, 1998



Interpreting the data

- 1 What are the two alleles for eye-colour of *Drosophila* in this data?
- 2 Which eye-colour is dominant? Explain.
- 3 What are the approximate phenotype ratios of F1 and F2?
- 4 Describe how eye-colour is inherited differently for male and female *Drosophila*.
- 5 What evidence in the data suggests that inheritance of eye-colour in *Drosophila* is sex-linked?
- 6 What are the most likely genotypes of the P1 parents? Explain.
- 7 Are the F1 parents homozygous or heterozygous? Explain.
- 8 Draw punnett squares showing that inheritance of eye-colour in *Drosophila* is sex-linked.
- 9 Under what conditions might a white-eyed female be produced? Explain using a punnett square diagram.
- 10 Is it possible for a male to carry the gene for white-eyes but have red eyes? Explain your answer.

Adaptation of the Wompoo Fruit Dove to environmental conditions

Introduction

The Wompoo Fruit Dove (*Ptilinopus magnificus*) is a species of bird that lives in the rainforests of eastern Australia and New Guinea. Scientists have named several subspecies (the name of a subspecies is the third word in its scientific name) of Wompoo Fruit Dove from different locations. These subspecies vary only slightly from one another in their markings but vary markedly in size. All subspecies of the Wompoo Fruit Dove are endangered, largely due to the destruction of their rainforest habitat.

Rainforests exist only in warm places where rainfall and humidity are high all year round. Rainforests are very diverse places that support a large number of species in their moist, humid environment. Wompoo Fruit Doves eat the abundant fruit in the rainforest and rarely move more than a few kilometres away from their place of birth in their lifetime.

Aim

To identify the trend in the size of Wompoo Fruit Doves from different locations and explain how this trend may have evolved.

Data

The subspecies of the Wompoo fruit dove (*Ptilinopus magnificus*)

Note: The sizes of the wing and the tarsus (part of the foot) are a reliable indicator of the size of the whole bird.

Ptilinopus magnificus assimilis—Cape York in far north Queensland

average wing length: 182.1 cm average tarsus length 26.2 cm

Ptilinopus magnificus magnificus—Northern NSW and north to Rockhampton in Queensland

average wing length 224.7 cm average tarsus length 33.9 cm

Ptilinopus magnificus poliura—South-eastern New Guinea

average wing length 164.1 cm average tarsus length 24.9 cm

Ptilinopus magnificus kerri—North Queensland around Cairns

average wing length 198.3 cm average tarsus length 28.5 cm

Ptilinopus magnificus sepestrionalis—Northern New Guinea

average wing length 160.6 cm average tarsus length 23.3 cm

Reference

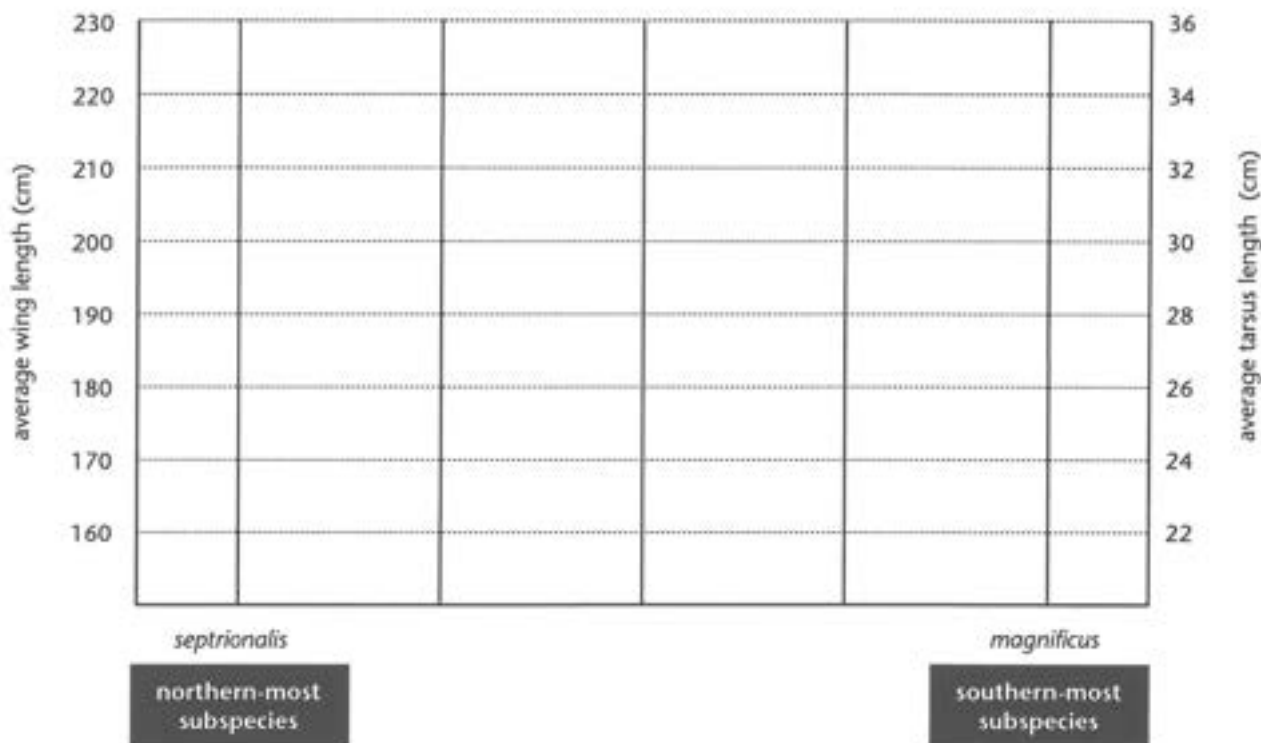
Frith, H. J. *Pigeons and Doves of Australia*, Rigby Publishing, Adelaide, 1982.

Organising the data

- 1 Firstly, arrange the subspecies in order of their location from north to south. Then complete the table below using information about each subspecies from the box above. The first and last rows are done for you.

Location	Subspecies name	Average wing length	Average tarsus length
Northern New Guinea	<i>sepestrionalis</i>	160.6 cm	23.3 cm
Northern NSW to Rockhampton in Queensland	<i>magnificus</i>	224.7 cm	33.9 cm

- 2 Graph the average wing and tarsus measurements for each subspecies in the same order as the table on the previous page.



- 3 Describe the trend in the size of Wompoo Fruit Doves in order of their location from north to south.

- 4 Circle the one environmental factor below that would vary progressively from north to south in the rainforests in which the Wompoo Fruit Doves live. (Remember the conditions needed for all rainforests!)

- ♦ predators
- ♦ rainfall
- ♦ temperature
- ♦ humidity
- ♦ availability of food

- 5 Explain how your chosen environmental factor would vary from north to south.

- 6 What advantage does the average size of each subspecies of Wompoo Fruit Dove offer in its particular location?

- 7 Explain how the environmental factor you have chosen may have caused the difference in the size of Wompoo Fruit Doves from different locations.

Using amino acid sequences to define evolutionary relationships

Introduction

Classification in biology is largely based on the structural characteristics of organisms. Organisms with similar structures are placed in the same group as it is assumed that they are closely related to each other and share a relatively recent common ancestor. However, it is often unclear which structural characteristics are more important from an evolutionary point of view. Also, comparing structural characteristics is not generally helpful when species are very different. The study of the biochemical similarities between organisms has recently become possible and has provided new evidence about evolutionary relationships.

Biochemical studies have made it possible to compare the biochemicals (such as proteins and nucleic acids) in various species. The degree of difference between the same protein or nucleic acid in different species provides a measure of the length of time since the two species shared a common ancestor. This is because mutations occur relatively regularly. So, the longer it is since a common ancestor, the greater the difference between molecules. It is this information that can be used in conjunction with fossils and other evidence to piece together evolutionary relationships.

Cytochrome C is a protein that first appeared in ancient bacteria about two billion years ago. It is an essential component of the biochemical pathway of aerobic respiration and is present in all modern organisms except anaerobic bacteria. Cytochrome C molecules each contain approximately 100 amino acids and the exact sequence of those amino acids is known for about 60 species.

Aim

To analyse a phylogeny (evolutionary tree) based on changes in Cytochrome C.

Data

See next page.

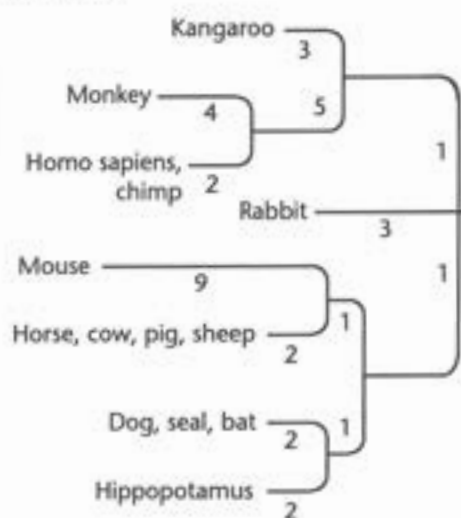
Reading the data

- 1 Draw a circle around all of the vertebrates (animals with backbones) in the diagram.
- 2 How many amino acid differences are there in the Cytochrome C of horses and pigs?
- 3 How many amino acid changes have taken place in the Cytochrome C of humans since the common ancestor of mice and people?
- 4 How many amino acid changes have taken place in the Cytochrome C of starfish since they shared a common ancestor with insects?
- 5 How many amino acid changes have taken place in the Cytochrome C of honey bees since they shared a common ancestor with plants?

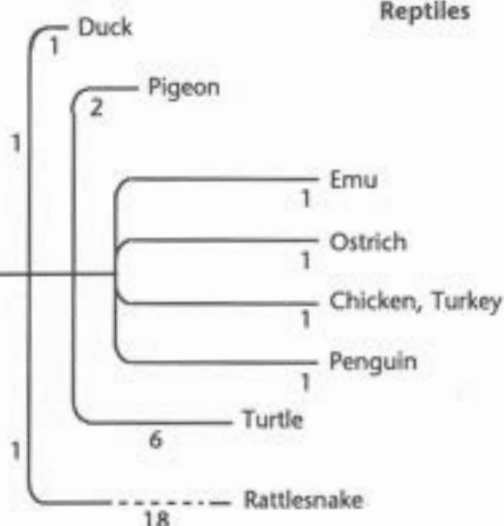
Interpreting the data

- 6 Use the diagram to decide which of the following statements are true. The first one is done for you.
 - a There are no differences in the amino acid sequences in Cytochrome C of humans (*Homo sapiens*) and chimpanzees. ☒
 - b There are six differences between the amino acid sequences in Cytochrome C of chimpanzees and monkeys. ☐
 - c Fungi, animals and plants all share a common ancestor. ☐
 - d Horses are more closely related to dogs than they are to sheep. ☐
 - e Vertebrates are more closely related to earthworms than they are to insects. ☐
 - f Rattlesnakes are more closely related to turtles than they are to ducks. ☐
 - g Birds share a more recent common ancestor than mammals. ☐

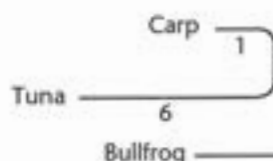
Mammals



Birds and Reptiles



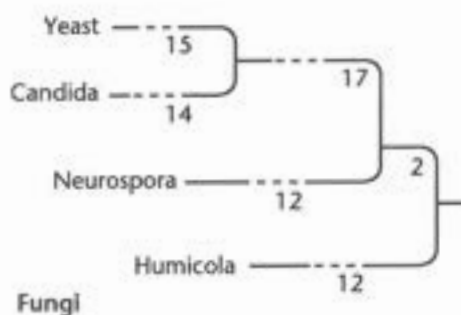
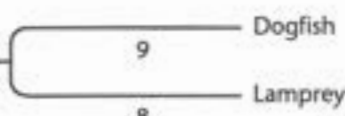
Bony fish



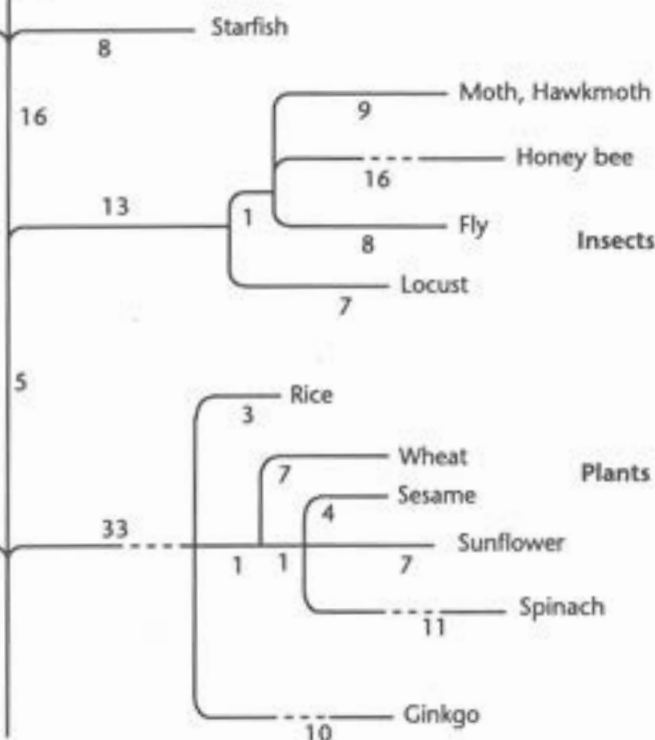
Amphibians



Cartilaginous fish



Fungi



Insects

Plants

(Curtis et al 1989)

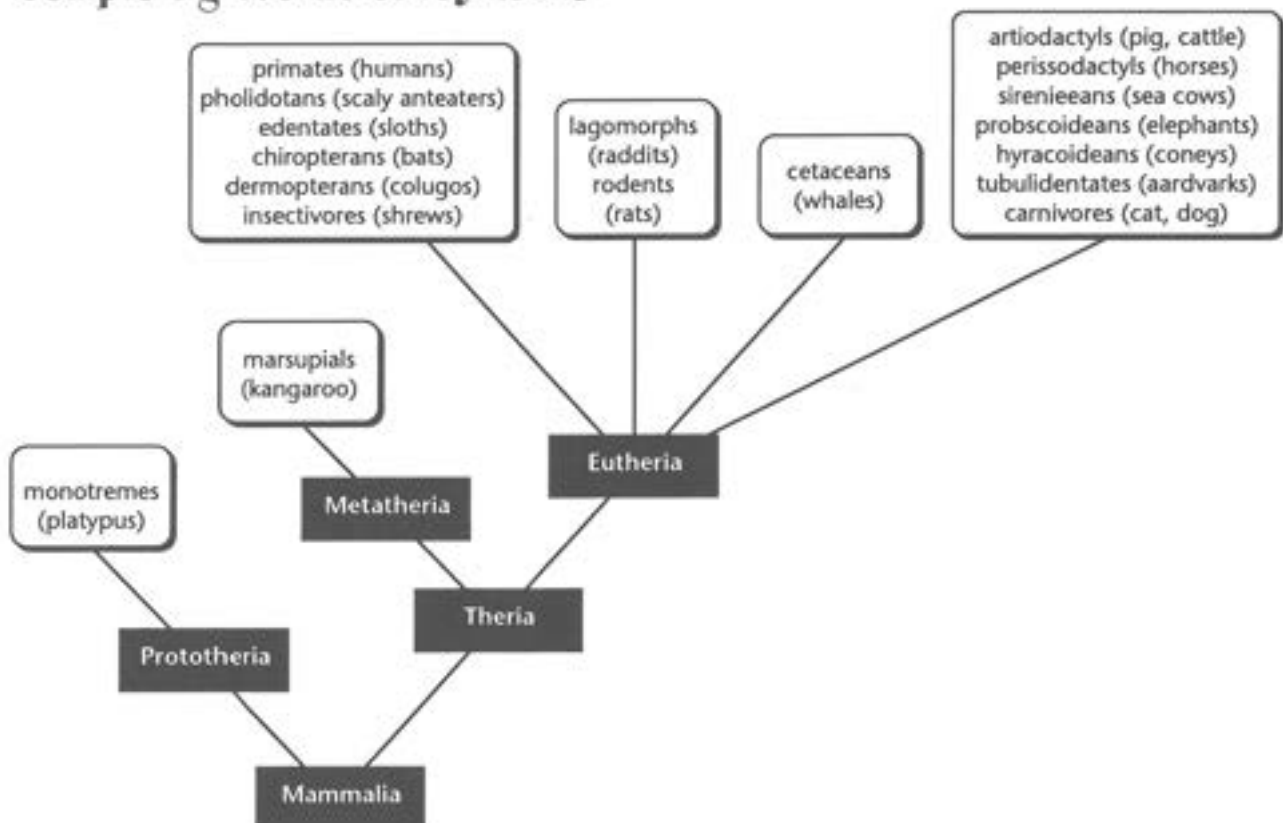
This is an evolutionary tree constructed using the amino acid sequences in the Cytochrome C of each organism. The numbers indicate the number of amino acids by which each Cytochrome C differs from the Cytochrome C at each branch point. Dashes indicate that a line has been shortened and is therefore not to scale. The ancestral Cytochrome C at each branch point is common to all organisms shown above it.

7 Change one or two words in each of the false statements from 6 to make them true.

8 For each true statement from 6 (including the ones that were changed in 7), outline the data from the diagram that supports it.

- a
b
c
d
e
f
g

Comparing evolutionary trees



The orders of mammals, arranged in evolutionary groupings based on their structural characteristics.

9 Compare the diagram above with the mammals' section of the Cytochrome C diagram. List one difference and one similarity in the way mammals are grouped between the two diagrams.

References

- Arms, K. *Biology*, CBS College Publications, USA, 1987.
Curtis, H. and Barnes, N. *Biology*, Worth publishing, USA, 1989.
Weisz, P. B. *The science of biology*, McGraw Hill, New York, 1971.

The mutagenic nature of Agent Orange

Introduction

Mutagens are substances that can cause permanent mutations. Mutations are changes to DNA: the genetic material responsible for controlling reproduction and the development of life. Once a mutation has occurred in a strand of DNA, the mutation will be replicated every time the cell divides. Mutations may be beneficial, harmful or may have no obvious effect. One of the harmful effects of mutations is the formation of some cancer cells. Mutations do occur naturally, but mutagenic substances make them occur more frequently.

During the Vietnam War in the late 1960s, the US military sprayed an estimated 95 million litres of Agent Orange (a herbicide mix) on about 10 percent of the area of southern Vietnam. Agent Orange has since been found to contain significant amounts of the chemical dioxin: a known mutagen. Many Vietnam War veterans have subsequently suffered from a range of medical problems that have been attributed to their exposure to Agent Orange.

Aim

To investigate the mutagenic nature of Agent Orange.

Data

Agent Orange is the code name for a herbicide developed for the United States military which was primarily used in tropical climates. Although the genesis of the product goes back to the 1940s, serious testing for military applications did not begin until the early 1960s. Agent Orange is a 50:50 mix of two chemicals:

- ◆ 2,4-dichlorophenoxy acetic acid (2,4-D) and
- ◆ 2,4,5-trichlorophenoxy acetic acid (2,4,5-T)

The combined product was mixed with kerosene or diesel fuel and sprayed from aircraft, vehicles and by hand.

Agent Orange was effective in defoliating (killing the leaves) and ultimately killing broad-leaf trees and shrubs in the dense jungles in Southeast Asia. The goal of the largest military defoliation campaign in history was to deny the enemy food or cover in dense forests.

The ecological effects of the spraying were sudden and severe: one spraying of Agent Orange was deadly to the enormous mangrove forests that protected the coast of South Vietnam and provided breeding and nursery quarters for large numbers of river and ocean fish. Inland, vast forests were replaced by a persistent grassland that burns frequently, killing tree seedlings.

The health effects of Agent Orange have received more attention than the ecological effects due to its contamination with TCDD or dioxin. Dioxin is considered one of the most toxic chemicals known. It was present in quite large quantities in many of the shipments of Agent Orange that went to Vietnam. How did dioxin exposure affect the millions of soldiers who served in Vietnam? The US Institute of Medicine set up a body called the Committee to Review the Health Effects in Vietnam Veterans of Exposure to Herbicides. The Committee found that:

- ◆ there is strong evidence that dioxin causes the skin disease chloracne and three cancers: soft-tissue sarcoma, non-Hodgkin's lymphoma and Hodgkin's disease.
- ◆ there is evidence that suggests links between Agent Orange and three other cancers: respiratory cancer, prostate cancer and multiple myeloma.
- ◆ Agent Orange may have other health effects including immune disorders, lowered sperm counts, diabetes, malformations and other reproductive and developmental effects and endocrine disruption.
- ◆ researchers have documented reproductive damage including cancer, spina bifida and immune problems in offspring of people exposed to dioxin.

References

National Institute for Science Education, 'Agent Orange Revisited', <http://whyfiles.org/025chem-weap/dioxin.html>

Interpreting the data

- 1 What does mutagenic mean? _____

- 2
 - a What is Agent Orange? _____
 - b When was Agent Orange used? _____
 - c Why was it used? _____
 - d When did it stop being used? _____
 - e Where was it used? _____
 - f How was it dispersed? _____
- 3 Do you think that people would have come into contact with the Agent Orange that was sprayed in their area? _____

- 4 What two chemicals make up Agent Orange? _____

- 5 What chemical contamination raised health concerns? _____

- 6 What were some ecological effects of the spraying of Agent Orange in South Vietnam? _____

- 7 What are *three* cancers caused by dioxin? _____

- 8 Name the other cancers that have been linked to Agent Orange. _____

- 9 What are *three* other possible health effects of Agent Orange? _____

- 10 Does the damage due to Agent Orange appear to stop with the first generation? _____

The development of antibiotic resistance in bacteria

Introduction

The discovery and development of antibiotics is one of the most important advances in the treatment of disease. Antibiotics are chemical substances that are capable of destroying or disabling bacteria. Where antibiotics are available, it is relatively rare for people to die from bacterial infections. However, strains of bacteria have appeared which are resistant to certain antibiotics.

Streptomycin is an antibiotic used to treat a range of bacterial infections including tuberculosis, pneumonia and urinary tract infections. A common cause of urinary tract infections is *Escherichia coli* (*E. coli*), a bacterium that is easily grown in culture. Initially, Streptomycin was very effective in killing the *E. coli* strains that caused urinary tract infections, so it was widely used as a treatment. Now there are several strains of *E. coli* that are resistant to Streptomycin.

For some time there were two possible explanations of how the original resistant bacteria arose.

Explanation 1: Bacteria that were resistant to Streptomycin were already present in the population and appeared due to chance mutation.

Explanation 2: The presence of Streptomycin in the environment caused the bacteria to mutate and become resistant.

Aim

To analyse experimental data to determine the origin of strains of *E. coli* bacteria that are resistant to Streptomycin antibiotic.

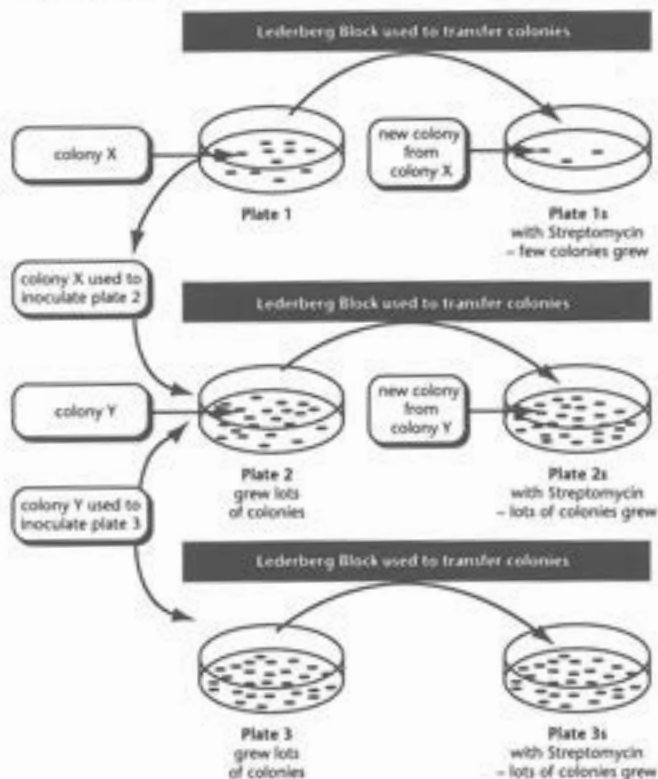
Data

An American biologist, J. Lederberg, devised a simple method of using a velvet covered block (Lederberg Block) to transfer bacterial colonies from one plate to another so that they ended up in exactly the same position on the new plate. The Lederberg method was used in the following experiment.

AIM: to determine whether exposure to Streptomycin causes *E. coli* to become resistant.

PROCEDURE

- 1 Prepare 6 nutrient agar plates and add Streptomycin to 3 of them.
- 2 Label the three plates without Streptomycin 1, 2 and 3 and the three with Streptomycin 1s, 2s and 3s.
- 3 Inoculate Plate 1 with *E. coli* and incubate.
- 4 Transfer colonies that grow on Plate 1 to Plate 1s using the Lederberg Block and incubate.
- 5 Use the location of colonies that grow on Plate 1s to identify a colony (X) from Plate 1 that contains resistant *E. coli*.
- 6 Remove colony X from Plate 1, dilute it and use it to inoculate Plate 2 then incubate.
- 7 Repeat steps 2 and 3 for Plate 2 and Plate 2s.
- 8 Repeat steps 2 and 3 for Plate 3 and Plate 3s.



RESULTS

Label	Plate 1	Plate 1s	Plate 2	Plate 2s	Plate 3	Plate 3s
Culture medium	Nutrient only	Nutrient plus Streptomycin	Nutrient only	Nutrient plus Streptomycin	Nutrient only	Nutrient plus Streptomycin
Resulting bacterial growth	Numerous colonies	Very few colonies	Many colonies	Many colonies	Many colonies	Many colonies

Interpreting the data

- 1 What was the purpose of using the nutrient agar plates with and without Streptomycin?
- 2 Explain why Plate 1 grew many colonies and Plate 1s grew very few colonies.
- 3 Is explanation 1 or 2 supported by the results in Plate 2 and Plate 2s? Discuss.
- 4 What result would have supported the other explanation?
- 5 Was the data from Plate 3 and Plate 3s necessary? Did it provide any further evidence to support either explanation?
- 6 Did the bacteria in Plate 3 come into contact with the Streptomycin at any stage during the experiment? Where did the resistant bacteria come from in Plate 3?
- 7 Which explanation does this experiment support? Explain your answer.
- 8 Are there any other possible explanations for the development of resistance to Streptomycin?
- 9 What further experiments would you suggest to explore alternative explanations?
- 10 The sentences in the boxes below make up a paragraph explaining how antibiotic resistance develops. Use arrows to show the most appropriate sequence. One arrow is done for you.

Research into how bacteria become resistant to antibiotics involves the application of knowledge about natural selection.

The use of an antibiotic causes a change in the environment that applies selection pressure on the population of bacteria.

Eventually, most of the bacteria in the population will be resistant to the antibiotic.

If the selection pressure remains because antibiotics continue to be used, the resistant bacteria will become more common with successive generations.

If there are any resistant bacteria in the original population, they will survive.

Bacteria susceptible to the antibiotic will die.

Using pedigrees to determine the inheritance of genetic diseases

Introduction

Genetic diseases occur in most human populations and arise due to mutations that create a deleterious (harmful) gene in one individual. If the deleterious gene is dominant, it will die out if it prevents people from reaching a breeding age. However, if it is recessive, it may remain in a population and get carried by heterozygous people and only appear in a homozygous individual.

Pedigrees can be used to determine whether diseases that seem to run in families are genetically inherited. By looking at a pedigree of a family with a history of a genetic disease, you can identify patterns of inheritance from which you can determine whether it is dominant or recessive. You can also use pedigrees to predict the likelihood that the children of two individuals will have the genetic disease. This information can be used in genetic counselling where partners can determine the risk of a genetically inherited disease appearing in their offspring.

Pedigree diagrams usually use the following symbols:

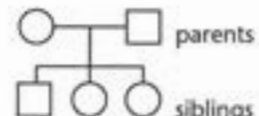
□ male without disease

■ male with disease

○ female without disease

○ female without disease

● female with disease

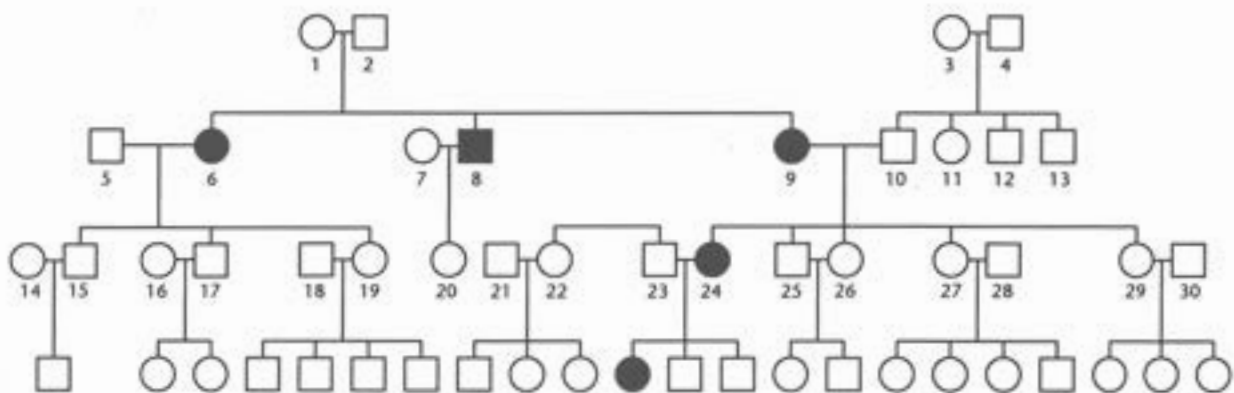


Aim

To determine whether genes for genetic diseases are dominant or recessive, to identify genotypes and predict offspring of particular crosses.

PEDIGREE 1: INHERITANCE OF FAMILIAL MEDITERRANEAN FEVER (FMF)

FMF is a genetic disease in which people experience repeated episodes of abdominal or chest pain, fever and leukocytosis (increase in white blood cells [leukocytes] in the blood).



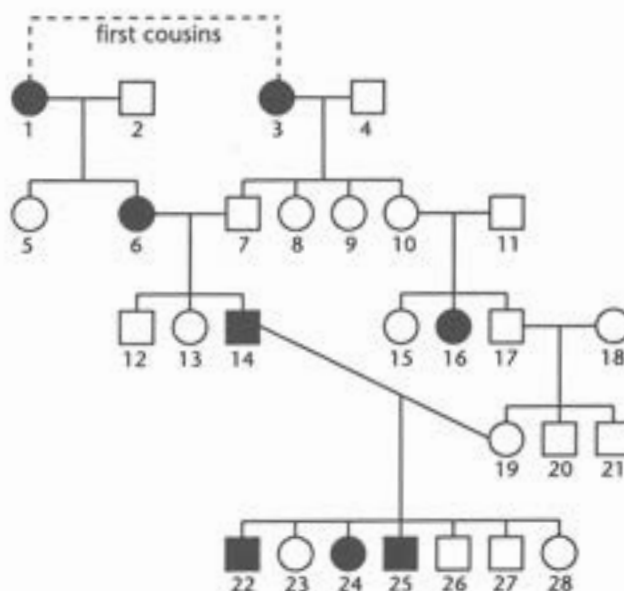
- 1 What is the relationship between:
 - a 1 and 2
 - b 2 and 9
 - c 3 and 9
 - d 15 and 16
 - e 15 and 17
- 2 How many children of each gender did 1 and 2 have?
- 3 Is FMF dominant or recessive? Explain why, using a specific cross from the pedigree.
- 4 Is it possible for 1 and 2 to have any normal children? Show your working.

- 5 Complete the table below using information that you worked out from the pedigree.

Number	Phenotype	Genotype	Explanation
1			
5			
6			
9			
10			
25			

PEDIGREE 2—INHERITANCE OF ALKAPTONURIA

Alkaptonuria is a rare genetic disease that causes the formation of coloured cartilage, dark urine and arthritis.



- What is the relationship between
 - 1 and 3
 - 14 and 17
 - 14 and 19
- How many children of each gender did 14 and 19 have?
- Is alkaptonuria dominant or recessive? Explain why, using specific crosses from the pedigree.
- Inbreeding is when closely related people have children. What are the two examples of inbreeding in this pedigree?
- What was the effect of inbreeding on the family of 14 and 19? Show how it occurred using a punnett square.
- Complete the table below using information that you worked out from the pedigree.

Number	Phenotype	Genotype	Explanation
1			
5			

The smallpox story

Introduction

'Smallpox was always present, filling the churchyard with corpses, tormenting with constant fear all whom it had not yet stricken, leaving on those whose lives it spared the hideous traces of its power, turning the babe into a changeling at which the mother shuddered, and making the eyes and cheeks of the betrothed maiden objects of horror to the lover.'

Macaulay 1800

Smallpox is a viral disease spread by droplet infection that has killed millions of people in epidemics that started thousands of years before Christ. The world's first vaccine was developed for smallpox and the disease is now almost totally eradicated.

Variolation is an early form of vaccination. It involves taking some fluid from a person suffering from a mild bout of smallpox and rubbing it into scratches on a healthy person. Following variolation, a smallpox rash usually appeared and, in some cases, became a serious bout of smallpox. However, it was widely believed that variolation gave some immunity to smallpox.

Aim

To explore the history of smallpox on Earth.

Data—Smallpox timeline

10,000 BC to

1000 BC—Smallpox probably first appeared in agricultural settlements in Africa; smallpox lesions appear on Egyptian mummies; first recorded smallpox epidemic during Egyptian/Hittite war in 1350 BC.

430 BC—Thucydides observed in Athens that those once infected with smallpox became immune.

180 AD—Massive epidemic probably contributed to the decline of the Roman Empire.

1500s—Smallpox decimated populations of North, Central and South America. Mexico's population fell from 25 to 1.6 million.

1600s—Variolation practised in India and China. First oral vaccination in China using pills made of fleas from cows. Variolation widely practised in Ottoman Empire (eg. Turkey).

1700s—Smallpox spread throughout the world, killing thousands, especially the native populations of colonised countries. Smallpox killed 400,000 people each year and a third of all survivors in Europe in the late 1700s went blind.

1718—First variolation performed by a doctor in England—Lady Montague ordered surgeons to perform it on her family. After testing variolation on prisoners, the Royal Family were variolated.

1722—First application of statistics to medical research: 1 in 14 children died after contracting smallpox but in children inoculated by variolation, only 1 in 91 died.

1765—It was widely accepted by rural communities that people who had been affected by cowpox did not contract smallpox and had no reaction to variolation. Doctors did not consider this significant.

1774—First vaccination: Benjamin Jesty, a farmer, vaccinated his children with material taken from the udder of a cow with cowpox by scratching their arms and applying the material. His wife nearly lost her arm due to inflammation and Jesty was ostracised by his community. However, the family did not contract smallpox during subsequent epidemics.

1791—Dutch Schoolmaster Peter Plett inoculated his employers and 2 children with material from cows with cowpox. He was discouraged because one child suffered a severe arm inflammation. However, the children he inoculated were the only people in the town to survive a smallpox epidemic 3 years later.

- 1796**—Edward Jenner extracted fluid from people with cowpox and inoculated a healthy 8-year-old boy who did not respond to variolation. Jenner reported this and 13 documented cases of immunity to smallpox following cowpox infection. Sir Joseph Banks of the Royal Society rejected the paper.
- 1798**—Jenner published a paper himself documenting his experiments and observations and coined the phrase 'vaccinae'. The medical establishment were sceptical but a number of doctors took up the practice.
- 1800**—Vaccination by cowpox pustule fluid was widespread with 100 000 people vaccinated world-wide.
- 1801**—US President Thomas Jefferson started a program of vaccinating American Indians, including the last of the Mohicans. Vaccine was adopted by all the major powers, including Napoleon, over the next decade.
- 1802**—The vaccine was so successful, Jenner predicted that smallpox would ultimately be eradicated.
- 1977**—The last case in the world of someone having caught smallpox from another person occurred in Somali.
- 1980**—Smallpox declared to be totally eradicated by the World Health Organization (WHO). The only surviving smallpox strains are held in high-security facilities in USA and Russia.
- 1996**—WHO recommended that final stocks of the virus be destroyed.
- 1999**—Deadline for the destruction of the final stocks of the smallpox virus.

References

Barquet, N and Domingo, P. Smallpox: the triumph over the most terrible of the ministers of death, *Annals of Internal Medicine*, 127:635-642, 1997. (<http://38.232.17.254/journals/annals/15Oct97/smallpox.htm>)

Macaulay, T. B. *The history of England from the Accession to James II*, Claxton, Remsen & Haffelfinger, Philadelphia, 1800.

Interpreting the data

- 1** When was the first recorded smallpox epidemic?
- 2** When and where was immunity to smallpox first observed?
- 3** Where was the first recorded type of vaccination practised and what was it called?
- 4** Describe the extent of the suffering from smallpox up until the 1700s.
- 5** Describe how the practice of variolation was introduced into England.
- 6** What was the first documented use of statistics in medical science?
- 7** Describe the link between smallpox and cowpox. Who first discovered the link?
- 8** Who were the first two people to use cowpox for a vaccination?
- 9** What was Jenner's contribution to the development of the smallpox vaccine?
- 10** Why do you think the Royal Society rejected the first paper by Jenner?
- 11** When was the term 'vaccine' first used?
- 12** What are the arguments for and against destroying the last remaining samples of the smallpox virus?

The quality of drinking water

Introduction

Providing water to any population of people is critical. It is even more critical when a big city is involved. In Sydney, for example, *Sydney Water* provides approximately 1500 million litres of water each day to more than 3.7 million people. An integrated network of 15 dams and storage reservoirs with associated catchments, eight water filters, more than 20,000 kilometres of water mains and canals, 162 pumping stations and 262 service reservoirs are used to gather, store and transport water. In addition to supplying a large quantity of water, they must also supply a high quality of water.

Aim

To understand how drinking water is monitored and treated to ensure that it is safe for human consumption.

Data

Raw water

Raw water is groundwater and water in rivers, streams and lakes that has not been treated to ensure that it is suitable for human consumption. Such waters can contain contaminants such as sediment, salts, organic material and pathogens such as *Giardia*. The level of treatment required to make the water suitable for human consumption depends on the overall condition of the catchment area (the area of land that catches the water and over which it flows to reach the storage place).

TREATMENT USING MULTIPLE BARRIERS APPROACH

1 *Catchment protection*

Waters from vegetated catchments dedicated to water supply are generally of a high quality, particularly if it is undisturbed by people and livestock. Water from these areas requires less treatment than those from catchments that have been used for other purposes such as agriculture, urban development or forestry. Waters from these disturbed catchments can be severely degraded and thus require more intensive levels of treatment. In addition, raw water from some sources may require pre-treatment to remove other contaminants.

2 *Storage detention*

Water is usually held in storage dams or tanks to allow sediments to settle to the bottom. If the water contains high concentrations of fine materials that stay in suspension, it may also be necessary to add chemicals such as aluminium or iron compounds which cause these materials to flocculate (stick together and settle to the bottom).

3 *Treatment and disinfection*

Water is often pre-treated by simple filtration to remove suspended sediments. Water supplied to most communities for drinking is then disinfected with chlorine or ultra-violet radiation so the risk of contamination by bacteria and other water-borne pathogens is minimised.

Monitoring

The quality of water supplied to homes is monitored and records are collected and publicised. Sydney water is tested for more than 70 different characteristics including taste, colour, odour, micro-organisms and chemical content. Microbiological data include:

- the percentages of samples that contain no coliform bacteria. Total coliform data are used as an indicator of the possible presence of disease causing organisms. The coliform bacteria are not disease-causing. Faecal coliforms come from human or animal faeces. 98% of samples should contain no faecal coliforms and 95% should contain no total coliforms.
- the amount of specific pathogenic organisms such as *Cryptosporidium*. This includes counting the number of *Cryptosporidium* oocytes (eggs) and the number of times doctors notify that people have *Cryptosporidium* infections.

Examples of data are given in Tables 1 and 2 below.

TABLE 1: PERCENTAGE OF WATER SAMPLES WITH NIL FAECAL COLIFORMS AND TOTAL COLIFORMS IN SYDNEY WATER SAMPLES

% of samples with nil coliforms	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998
Faecal coliforms	99.0	98.0	99.0	99.0	99.0	100.0	100.0	99.0	99.4	99.7
Total coliforms	82	85	96	93	89	92	92	94	95.7	98

TABLE 2: CRYPTOSPIRIDIUM NOTIFICATIONS AND OOCYTE COUNTS IN SYDNEY WATER SAMPLES

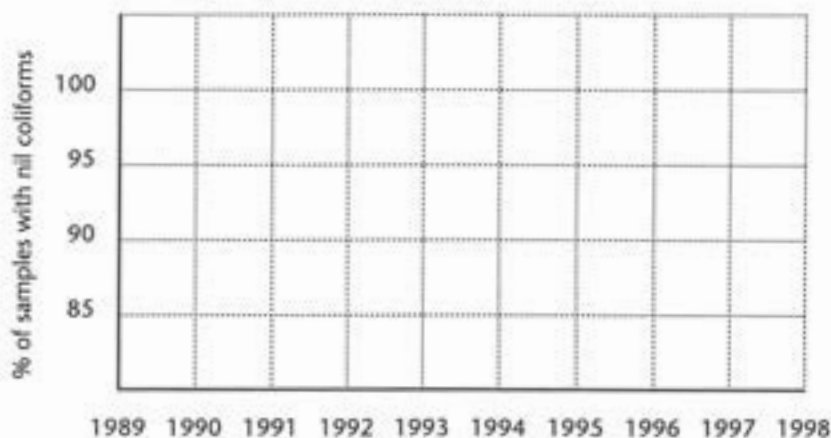
Year	97		98											
Month	N	D	J	F	M	A	M	J	J	A	S	O	N	D
Oocyte counts per test per week	0	0	0	0	0	0	0	0	0	80	300	0	0	0
Notifications per month	20	80	120	350	400	110	0	10	50	0	15	5	15	25

References

NSW Dept of Health, *The health of the people of New South Wales*, Report of the Chief Health Officer, 2000.
Sydney Water (<http://www.sydneywater.com.au/>)

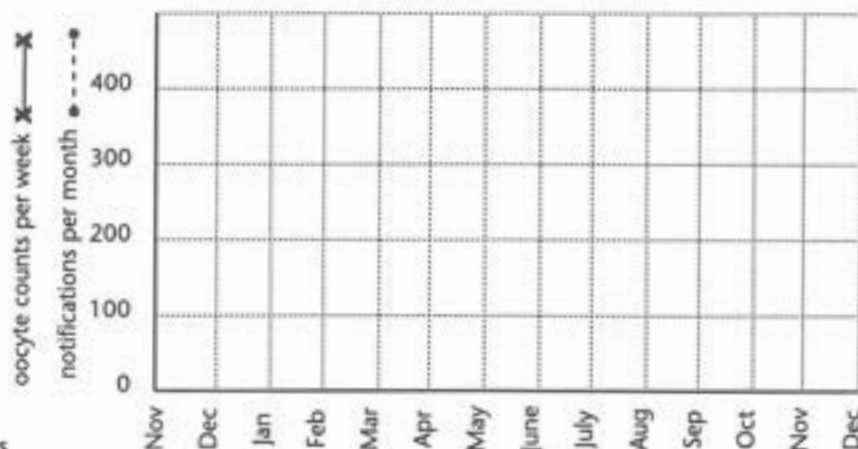
Describing the data

- Plot the data from Table 1 on the grid on the right.
- In what years did the water supply have above the recommended amount of coliforms?



- Plot the data from Table 2 on the grid on the right.

- The recommended allowable level of *Cryptosporidium* is zero. Predict which date the boiling of drinking water was recommended due to *Cryptosporidium* contamination.

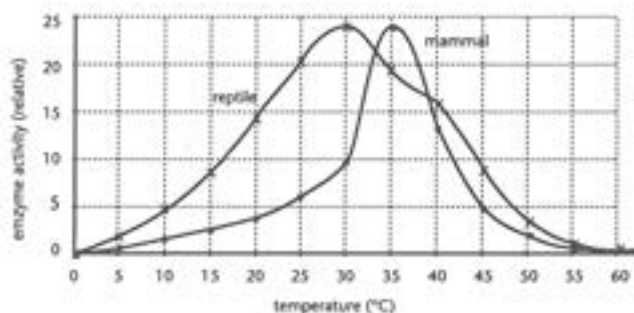


- What is a water catchment? What catchment characteristics cause the water from them to be degraded?
- How should catchments be managed to minimise the amount of water treatment that is needed to make water safe for drinking?
- What is flocculation and why is it done?
- What is water disinfection? What is used to disinfect water?

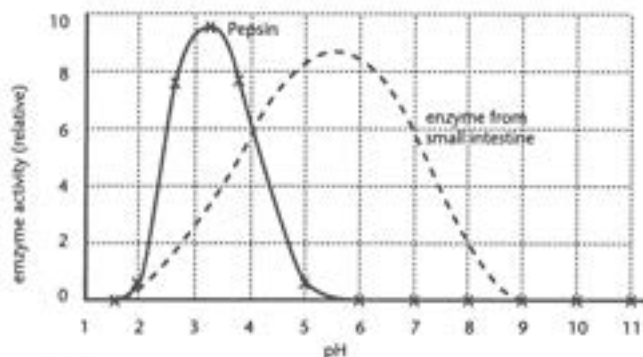
Selected answers

Activity 1: The effects of temperature and pH on enzyme activity

Part 1. The effect of temperature on the activity of the enzymes of endothermic and ectothermic animals



Part 3. Investigating the effect of pH on the enzyme activity



Part 2. Investigating the effect of temperature on enzyme activity

Treatment temperature (°C)	Milk only	Milk and renin
0	No change	No change
37	No change	Solidifies
80	No change	No change

Interpreting the data

Part 1

- Enzyme activity in mammals reaches a peak at 37°C but falls sharply either side of 37°C.
- 30°C to 40°C
- Enzyme activity in reptiles reaches a peak at 30°C and falls gently either side of 30°C.
- 20°C to 40°C
- Because mammals are endothermic and their bodies have evolved to stay at a temperature close to 37°C, whereas reptiles are ectothermic and their enzymes must be able to function well at a wider range of temperatures.

Part 2

- The action of renin on milk at 37°C has caused the milk solids to combine.
- It is a control. It shows that milk won't solidify with just heat.
- Temperature and the presence of renin
- Milk
- Renin is active at 37°C but not at 0°C or 80°C.
- Yes, but they would probably be active at a wider range of temperatures than enzymes in mammals because reptiles' body temperature fluctuates more because they are ectothermic.

Part 3

- The activity of pepsin rises to a peak at a pH of 3.5 then falls to zero at pH 6.
- Pepsin is most active at a pH between 3 and 4.
- Because the pH in the intestine is about 7 and the stomach is about 4 due to the hydrochloric acid secreted in the stomach.

Activity 2: Feedback control of blood sugar in humans

Interpreting the data

- Exertion or fasting. Exertion removes glucose from the blood for use in the muscles. When fasting, little glucose will be being absorbed into the blood.
- The alpha cells secrete glucagon into the blood.
- The liver produces glucose to increase blood glucose levels.
- The absorption of nutrients, including glucose, after a meal.
- Beta cells in the pancreas secrete insulin into the blood.
- Glucose transport to the cells is increased.
- The liver converts glucose into glycogen for storage which decreases the blood glucose level.
- 180 mg/100 ml
- More glucose is produced by the liver.
- Digestion, glycogenolysis and gluconeogenesis.
- It can be excreted in urine, used for fat formation, converted to glycogen in the liver by glycogenesis and used for respiration.

Making links

- High blood glucose
- Insulin causes blood glucose to fall.
- Low blood glucose
- Glucagon causes blood glucose to rise.
- Because the level of glucose stimulates hormonal changes that cause the level of glucose to change.

Activity 3: Products extracted from donated blood

Interpreting the data

- 1 If it is diseased.
- 2 In cases of emergency: when the patient will die without it.
- 3 Red and white blood cells and plasma
- 4 Red blood cells are used to increase the oxygen carrying capacity of the blood. White blood cells are used to treat serious infections or if white cell count is low.
- 5 The cryosupernatant that is extracted from the plasma.
- 6
 - a treating massive bleeding—cryoprecipitate
 - b patients who have a low white cell count—white blood cells
 - c controlling bleeding following a severe haemorrhage—platelets and cryoprecipitate
 - d managing Hepatitis B—Hyper immunoglobulin
 - e protection against chicken pox—Hyper immunoglobulin
 - f boosting the immune system after a marrow transplant—intragam
- 7 Platelets, cryoprecipitate, cryosupernatant
- 8 Centrifuge
- 9 Anti D, immunoglobulin, Intragam
- 10 To carry oxygen around the body
- 11 Draw lines to link each blood product with its use.

Blood product	Use
Whole blood	To boost the immune system after bone marrow transplants
Plasma	To control Rhesus disease in newborn babies
Red blood cells	In cases of serious infection or due to a very low white cell count
White blood cells	To prevent Hepatitis A
Cryoprecipitate	To increase the oxygen-carrying capacity of the blood for example, after bleeding
Hyper Immunoglobulin	When there is severe blood loss and patient will die without it
Intragam	To control bleeding due to deficiencies in the clotting of blood
Normal Immunoglobulin	For protection against a variety of viral diseases
Anti D	Most commonly used in the treatment of massive bleeding
Platelets	To quickly increase blood proteins

Activity 4: Temperature regulation in ectotherms and endotherms

Recording the data

Animal	Environment	Adaptations for heat control		
		Structural	Behavioural	Physiological
Thorny Devil	Desert	Thick, spiny skin	Flattens body and lies in the sun in morning, lifts body off sand in day, remains motionless for long periods	
Mitchell's Hopping Mouse	Desert	Large thin ears, thick fur, small slim body	In burrow in day, feeds at night, huddles when cold	Fluffs out its fur, ears have large blood supply and sweat though the day
Corroboree Frog	Alpine	Mostly black skin	Active on warm days, inactive through the winter and at night	Accumulates fat in summer
Mountain Pigmy Possum	Alpine	Thick fur, round stocky body, small ears with limited circulation	Rolling into a furry ball, hibernates in winter	Increases body weight through warmer months, slower metabolic rate in winter

Interpreting the data

- 1 The Mitchell's Hopping Mouse and the Mountain Pigmy Possum because they have fur for insulation and they are capable of moving quickly even in the cold.
- 2 The Thorny Devil and the Corroboree Frog because they have no fur for insulation and they slow down when it is cold.
- 3 In the morning, it lies flat on the sand to warm up. In the day, it raises itself off the sand to avoid overheating.
- 4 It remains in its burrow to avoid the heat and cools itself by sweating from its large ears.
- 5 It huddles to limit heat loss at night.
- 6 Because when the environment is cold, the frog is cold. As a result, its metabolism slows down and it is unable to move.
- 7 It reduces the surface area exposed to the cold and as a result reduces heat loss.
- 8 The large ears with sweat glands and good blood supply of the Mitchell's Hopping Mouse help it to lose heat in the desert. The Mountain Pigmy Possum has small ears to limit heat loss in the cold mountains.
- 9 To warm their bodies as quickly as possible so they are able to move quickly.
- 10 The temperature of the Mitchell's Hopping Mouse and the Mountain Pigmy Possum would remain constant, whereas the Corroboree Frog and the Thorny Devil would change temperature with the environment.

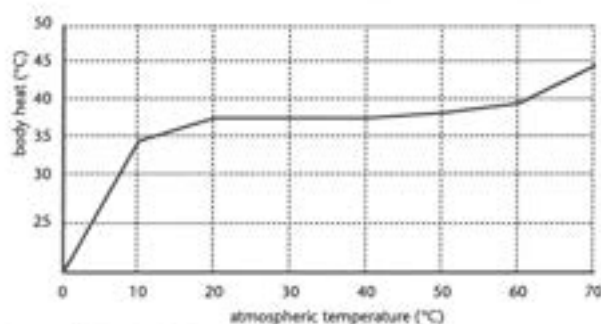
Making links

Adaptation	Reason or advantage
Large thin ears	Radiate heat into surrounding air
Lizard lying in the ground in the early morning	Avoid extremes in temperatures
Roll into a furry ball	To warm the body
Thick fur	To insulate, keeping heat inside body
Lizard burrowing	To prevent heat loss by reducing the SA/Vol

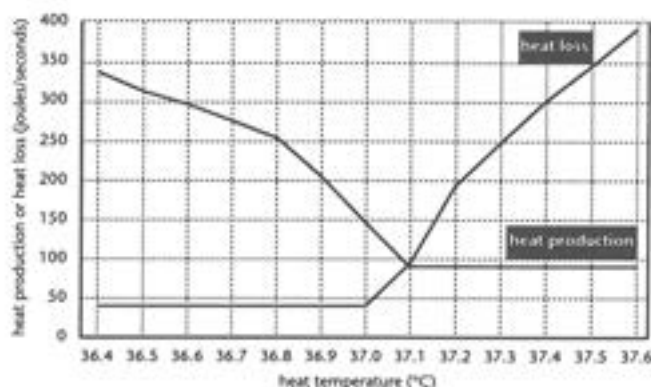
Activity 5: Regulation of body temperature in humans

Graphing the data

Body temperature at different atmospheric temperatures



Heat production and heat loss against head temperature



Describing the data

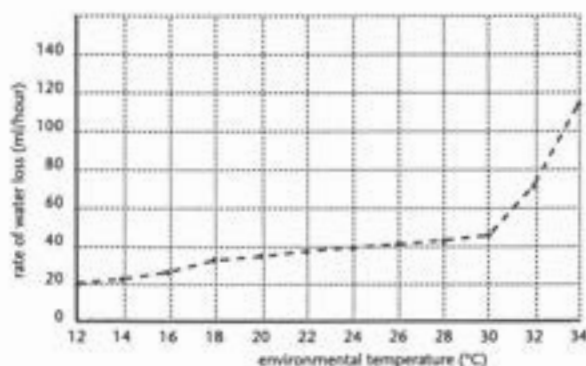
- Between 20°C and 50°C
- Perspiration and redness
- Cold skin, shivering, extremities and lips appear blue.
- Between 37°C and 37.1°C

5 Interpreting the data

Body response or behaviour	Physiological explanation
Sleepiness and disorientation	Person has hypothermia and their low body temperature is now disrupting the biochemistry of the brain.
Curled in a ball, arms crossed	These behaviours reduce the surface area of the body exposed to cold and so reduce the rate of heat loss.
Lips and nose appear blue	Peripheral blood vessels (those in the skin) are constricted (closed up) reducing the blood supply to the skin. This cools the skin and reduces the rate of heat loss.
Swinging arms, moving around	Body movement increases heat production by the body.
Skin feels cold	Perspiration evaporates from the skin causing evaporative cooling. Heat moves from the blood into the cool skin.
Shivering	Peripheral blood vessels (those in the skin) are constricted (closed up) reducing the blood supply to the skin. This cools the skin and reduces the rate of heat loss.
Perspiration	Perspiration evaporates from the skin causing evaporative cooling. Heat moves from the blood into the cool skin.
Arms and legs spread out	These behaviours increase the surface area of the body and so increase the rate of heat loss.
Redness in the face and skin	Peripheral blood vessels are dilated (opened up) increasing the blood supply to the skin, increasing heat loss.
Sweating diminishes	Due to prolonged exposure to heat, the body has become dehydrated and, as a result, is no longer able to perspire at the same rate.

- Dilation of peripheral blood vessels sends more blood to the skin to be cooled by perspiration. Spreading the arms and legs exposes the skin surface to the air for evaporative cooling. These adaptations all increase heat loss.
- Movement, shivering and increased metabolic rate increase heat production. Constriction of peripheral blood vessels reduces the volume of blood that flows through the cool skin and reduces heat loss.
- A very small change in head temperature stimulates metabolic changes. Other parts can fluctuate more in temperature.

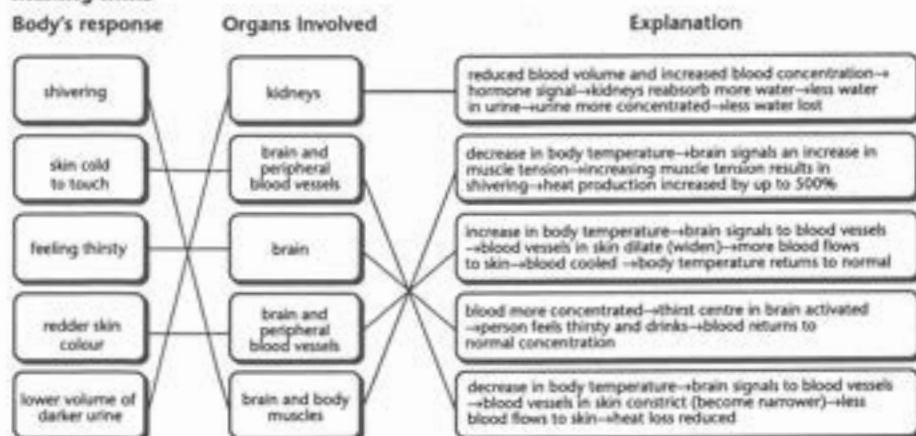
Activity 6: Temperature control and water loss in humans



Describing the data

- 1 Body temperature remains constant despite changes in environmental temperature.
- 2 Water loss increases as environmental temperature rises.
- 3 At about 28°C.
- 4 The body starts to sweat and the skin colour becomes redder.

Making links



- ◆ The peripheral blood vessels help control temperature by regulating the volume of blood passing through the skin: reducing the volume when it is cold to reduce heat loss and increasing the volume when it is hot to increase heat loss.
- ◆ Sweating causes evaporative cooling to increase heat loss. This also results in water loss.

Activity 7: The adaptations of mangroves

Interpreting the data

- 1 Plants need oxygen to respire.
- 2 They get oxygen from air spaces between soil particles.
- 3 They have above ground roots (pneumatophores) with lenticels which allow oxygen absorption from the air.
- 4 The seed needs to become buried in moist soil containing air spaces. This usually occurs on the ground surrounding the tree.
- 5 Grey Mangrove seeds are more likely to fall into the water or onto waterlogged soil than seeds from terrestrial plants.
- 6 The seeds germinate while they are still in the fruit on the plant and then float on the water until they make contact with land.
- 7 Soil water in terrestrial habitats is relatively low in salt. Water moves into the roots by osmosis because the salt concentration inside the roots is greater than in the water. The same thing happens in mangroves except the salt content in both the water and roots is greater than in terrestrial habitats.
- 8 Water would move out of the roots and into the soil, causing the plant to dehydrate.

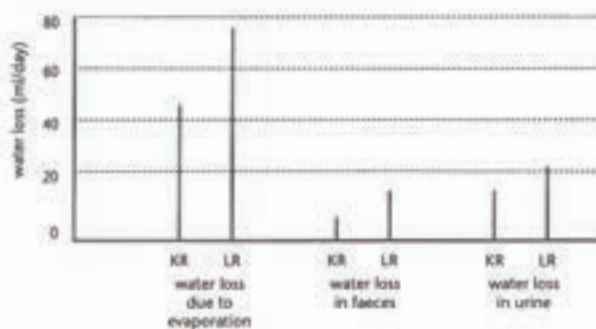
9 The prevention of salt accumulation in Grey Mangroves



Activity 8: Water balance in terrestrial environments

Water loss or gain	ml of water/kg body weight /day	
	Kangaroo rat	laboratory rat
Gain due to oxidation of carbohydrate, fat and protein	+54.0ml	+54.0ml
Loss due to evaporation	-44.0ml	-77.0ml
Loss in faeces	-2.5ml	-13.6ml
Loss in urine	-13.5ml	-22.0ml
Water balance (excluding diet)	-6.0ml	-48.6ml
Minimum water required in diet for water balance	6.0ml	48.6ml

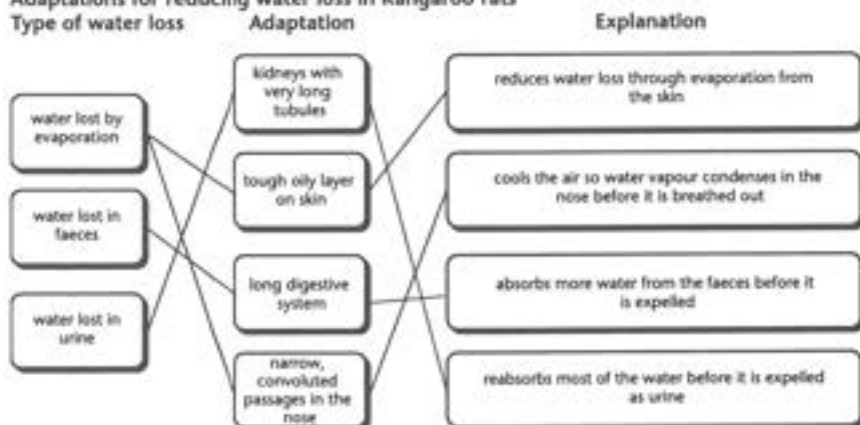
Bar graph of the components of water loss from a Kangaroo rat (KR) and a laboratory rat (LR)



Interpreting the data

- 1 By evaporation
- 2 Kangaroo rats lose much less water than laboratory rats.
- 3 The laboratory rat needs to ingest more water to maintain water balance because it loses more water.
- 4 The figure is calculated per kilogram of body weight and both are kept under the same dietary and environmental conditions.
- 5 The rats could avoid going out in the day.
- 6 Water loss would be less because the rate of evaporation is greatest during the day.

Adaptations for reducing water loss in Kangaroo rats



Kangaroo rats reduce water loss:

- due to evaporation by condensing water vapour in their narrow nose passages before they breathe it out.
- in faeces by absorbing more water from the faeces in their long digestive system.
- in urine by reabsorbing most of the water in their long kidney tubules.

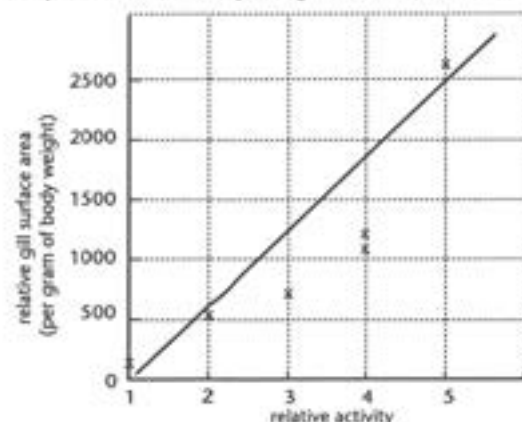
Activity 9: Gill surface area and level of activity in fish

Data

Type of fish	relative gill surface area (per gram of body weight)	habitat/habits	Relative level of activity (1=least active, 5=most active)
Puffer	505	Swims slowly around in shallow water feeding on corals and small crustaceans.	2
Sea Bass	1111	Leaps into the air when chasing prey in coastal waters but does not swim long distances	4
Sea Robin	712	Moves slowly around on the bottom of the ocean feeding on crustaceans	3
Sea Trout	1253	Lives in the open ocean and around the coast. Swims up rivers to breed	4
Goosefish	51	Moves slowly and rarely, remaining still on the ocean floor until prey comes within reach	1
Mackerel	2551	Lives, travels long distances in the open ocean, swimming very quickly to feed on prey	5

Processing the data

Graph of relative activity and gill surface area in fish



Interpreting the data

- 1 The gill surface area per gram of body weight increases as the level of activity in fishes increases.
- 2 More active fish need more oxygen: increased gill surface area allows more oxygen to be absorbed at a faster rate.
- 3 Warmer water holds less oxygen, so tropical fish may need a larger gill surface area.

Making links

- 4 Yes, because a high level of activity always means a high demand for oxygen.
- 5 The lung surface area per gram of bodyweight of endothermic animals would be greater than ectothermic animals because more energy is required to keep endothermic animals warm. This leads to a greater demand for oxygen.

- Oxygen is more available in air (20%) than in water (<1%). No, the gill surface area per gram of bodyweight of fish would be greater than the lung surface area per gram of bodyweight of terrestrial reptiles because there is a lower percentage of oxygen in water.
- Level of activity, temperature, availability of oxygen.

Activity 10: Inheritance of eye-colour in *Drosophila*—is it Mendelian?

Interpreting the data

- Red and white eyes
- Red eye-colour is dominant because the F1 generation is red-eyed.
- F1 are all red-eyed while F2 is made up of 2 red-eyed females, 1 red-eyed male and 1 white-eyed male.
- There are no white-eyed females in the F2 generation.
- The ratios are not the expected 3:1 ratio, and the characteristic is inherited differently for males and females.
- The female is homozygous for red eyes and the male has one gene for white eyes on the X chromosome.
- The female is heterozygous because she inherits a white-eyed gene from her father and a red-eyed gene from her mother. The male is neither heterozygous nor homozygous because he has only one red-eyed gene from his mother on the X chromosome.

8

	X^w	X
X	$X X^w$	$X X$
Y	$X^w Y$	$X Y$

Ratio of phenotypes
2 red-eyed female
1 red-eyed male
1 white-eyed male

9

When a white-eyed male & heterozygous female produce offspring.

	X^w	X
X^w	$X^w X^w$	$X^w X$
Y	$X^w Y$	$X Y$

Ratio of phenotypes
1 white-eyed female
1 red-eyed female
1 white-eyed male
1 white-eyed female

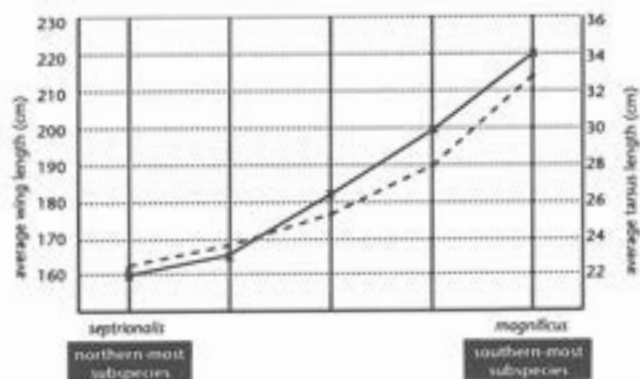
- No, because the male only has one gene for eye-colour. As a result, he cannot carry a gene for eye-colour.

Activity 11 - Adaptation of the Wompoo Fruit Dove to environmental conditions

Organising the data

Location	Subspecies name	Average wing length	Average tarsus length
Northern New Guinea	<i>septrionalis</i>	160.6 cm	23.3 cm
South eastern New Guinea	<i>pollura</i>	164.1 cm	24.9 cm
Cape York in far north Queensland	<i>assimilis</i>	182.1 cm	26.2 cm
North Queensland around Cairns	<i>keri</i>	198.3 cm	28.5 cm
Northern NSW to Rockhampton in Queensland	<i>magnificus</i>	224.7 cm	33.9 cm

2



- The size of Wompoo Fruit Doves increases from north to south.
- temperature
- Temperature would decrease from north to south.
- A larger body size is an adaptation to cooler climates and a smaller body size is an adaptation to warmer climates. As the temperature decreases, the body size gets bigger. This is due to surface area to volume ratio. Larger birds lose heat slowly because they have lower sa/v while smaller birds lose heat fast because they have a higher sa/v ratio.
- The original population of Wompoo Fruit Doves spread out to populate the rainforest areas from northern NSW to northern New Guinea. The different parts of the Wompoo population were relatively isolated from each other because of the birds' non-migratory behaviour and because the rainforests receded due to climatic changes. Through natural selection, larger birds were more likely to survive in cooler climates and smaller birds were more likely to survive in warmer climates. Over time, the separate populations have evolved different average sizes.

Activity 12: Using amino acid sequences to define evolutionary relationships

Reading the data

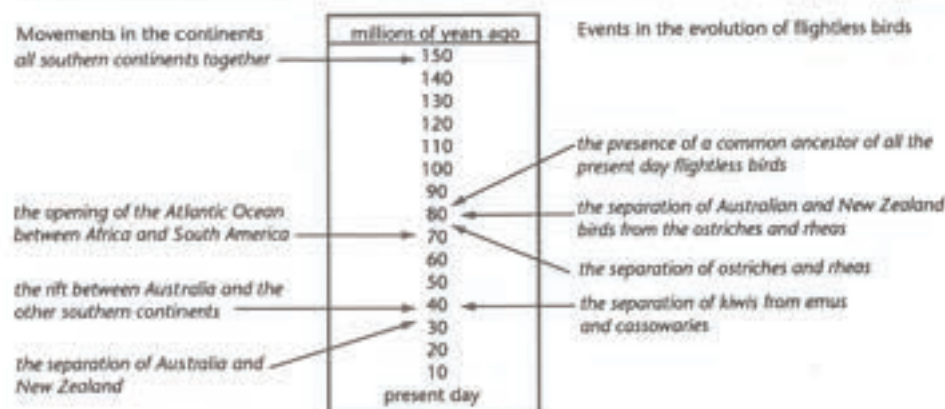
- None. They are identical.
- 7
- 24
- 35

Activity 15: Continental drift and the distribution and evolution of flightless birds

Describing the data

1

Event or evidence	When did it occur? (millions of years ago)
all southern continents were together	150
the rift between Australia and the other southern continents	39
the opening of the Atlantic Ocean between Africa and South America	65
the separation of Australia and New Zealand (not on the diagrams)	40
the presence of a common ancestor of all the present day flightless birds	90
the separation of Australian and New Zealand birds from the ostriches and rheas	80
the separation of ostriches and rheas	80
the separation of kiwis from emus and cassowaries	40

2 Movements in the continents
all southern continents together

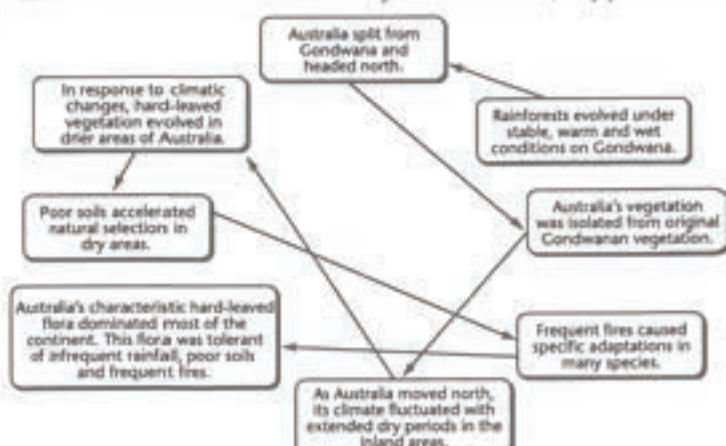
Making links

- All of the birds have similar DNA. The bigger the difference between 2 birds, the older their common ancestor is.
- It supports it, because all birds come from continents that used to be joined.
- Yes, the 2 continents began separating between 115 and 65 million years ago and DNA evidence suggests ostriches and rheas shared a common ancestor about 80 million years ago.
- Yes, kiwis and Australian birds shared a common ancestor about 45 million years ago and the 2 countries separated 40 million years ago.

Activity 16: The evolution of Australian plants

Questions

- Broad-leaved rainforest
- Rainforests of Gondwana
- Antarctica and South America
- Warm and wet conditions
- About 35 million years ago, when Australia split from Gondwana
- Cooler, more changeable climate with dry periods
- Rainforest contracted to areas around the coast which remained warm and wet. New species evolved in areas where climatic fluctuations occurred, such as species tolerant of dry periods.
- Both Australian and South American rainforests come from the same original forests and have experienced similar environmental conditions since.
- The vegetation in both continents has evolved for more than 35 million years under different environmental conditions. For example, Australia's environment is characterised by a variable climate, very poor soils and frequent bushfires.
- 47



5

Number	Phenotype	Genotype	Explanation
1	Normal	Ff	Must be Ff to have FMF child
5	Normal	Ff or FF	Must have at least one normal gene
6	FMF	ff	All FMF people are ff because it is recessive
9	FMF	ff	All FMF people are ff because it is recessive
10	Normal	Ff	Must be carrying the FMF gene
25	Normal	Ff	Must have one FMF gene from his mother

Pedigree 2 - Inheritance of alkaptonuria

- First cousins
 - First cousins
 - 14 is 19's Uncle, 19 is 14's niece
- 4 sons and 3 daughters
- Alkaptonuria is recessive because 10 and 11 are normal and they had an affected daughter (16).
- 6 and 7 (second cousins), 14 and 19 (uncle and niece)

- They had a large number of affected children because 14 is homozygous recessive and 19 must be heterozygous.

	(14)	
	a	a
(19)	A	Aa
	a	aa

Number	Phenotype	Genotype	Explanation
1	alkaptonuric	aa	Since the condition is recessive, it must be homozygous
5	normal	Aa	Must have received a recessive gene from her mother

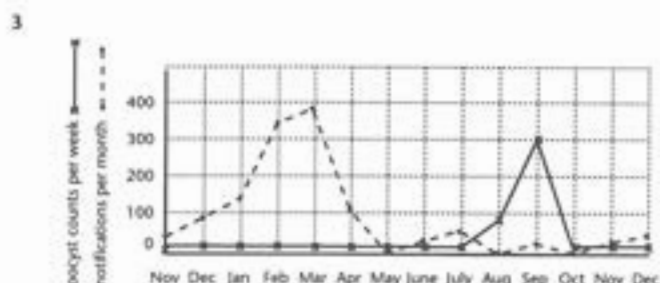
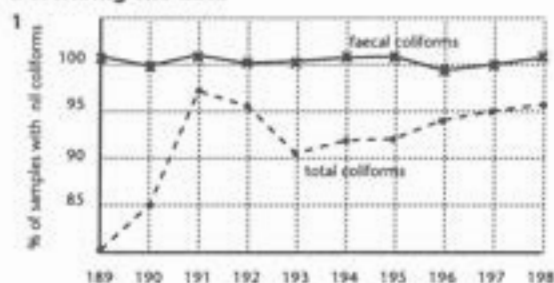
Activity 16: The smallpox story

Interpreting the data

- The first recorded smallpox epidemic was during the Egyptian/Hittite war in 1350 BC.
- Immunity was first observed in 430 BC in Greece. Those infected with smallpox who survived became immune.
- Variolation was first practised in India and China in the 1600s.
- Smallpox killed 400,000 people each year and a third of all survivors in Europe in the late 1700s went blind.
- Lady Montague ordered surgeons to perform it on her family.
- 1 in 14 children died after contracting smallpox but in children inoculated by variolation, only 1 in 91 died.
- People who had been affected by cowpox did not contract smallpox. This was widely accepted by rural communities.
- The first people to use cowpox as a vaccination were Benjamin Jesty (a farmer) and Peter Plett (a Dutch Schoolmaster).
- Jenner published a paper himself documenting his experiments and observations and coined the phrase 'vaccinae'.
- The paper was possibly rejected because rural people rather than doctors had discovered the vaccine.
- The term was first used by Jenner in 1798.
- Use this as an opportunity for class discussion.

Activity 17: The quality of drinking water

Describing the data



- 1989, 1990, 1992-1996.
- The boiling of drinking water was recommended in August 1998
- A water catchment is an area of land that collects water for a water supply and the land over which the water flows to reach the storage place. Water is degraded from catchments that have been used for other purposes such as agriculture, urban development or forestry.
- Catchments should not be disturbed by people and livestock.
- Flocculation is the adding of chemicals such as aluminium or iron compounds to cloudy water to make the suspended material stick together and settle to the bottom.
- Water disinfection is treating water with chlorine or ultra-violet radiation so that the risk of contamination by bacteria and other water borne pathogens is minimised.

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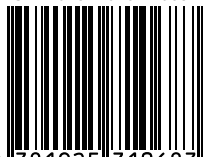
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